

Research papers

Simultaneous state-parameter estimation supports the evaluation of data assimilation performance and measurement design for soil-water-atmosphere-plant system

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ABSTRACT

Improvements to agricultural water and crop managements require detailed information on crop and soil states, and their evolution. Data assimilation provides an attractive way of obtaining these information by integrating measurements with model in a sequential manner. However, data assimilation for soil-water-atmosphere-plant (SWAP) system is still lack of comprehensive exploration due to a large number of variables and parameters in the system. In this study, simultaneous state-parameter estimation using ensemble Kalman filter (EnKF) was employed to evaluate the data assimilation performance and provide advice on measurement design for SWAP system. The results demonstrated that a proper selection of state vector is critical to effective data assimilation. Especially, updating the development stage was able to avoid the negative effect of “phenological shift”, which was caused by the contrasted phenological stage in different ensemble members. Simultaneous state-parameter estimation (SSPE) assimilation strategy outperformed updating-state-only (USO) assimilation strategy because of its ability to alleviate the inconsistency between model variables and parameters. However, the performance of SSPE assimilation strategy could deteriorate with an increasing number of uncertain parameters as a result of soil stratification and limited knowledge on crop parameters. In addition to the most easily available surface soil moisture (SSM) and leaf area index (LAI) measurements, deep soil moisture, grain yield or other auxiliary data were required to provide sufficient constraints on parameter estimation and to assure the data assimilation performance. This study provides an insight into the response of soil moisture and grain yield to data assimilation in SWAP system and is helpful for soil moisture movement and crop growth modeling and measurement design in practice.

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1. Introduction

The knowledge of soil water content and crop growth statuses is critical for agricultural management practices (MacDonald and Hall, 1980). However, in-situ and direct measurements of these information with adequate frequency and accuracy are very expensive and time-consuming. Various models have thus been developed to provide these information by simulating the interactions between soil, atmosphere, and plant, including WOFOST (Supit et al., 1994), DASSAT (Hoogenboom, 2004), EPIC (Sharpley and Williams, 1990), SPA (Sus et al., 2010), STICS (Brisson et al., 2003) and SWAP (van Dam et al., 1997), etc. The robustness of such models depends on their ability to imitate the crop biomass and yield in response to water, nitrogen, and other environmental fac-

tors, and relies on the availability of soil properties, crop parameters, meteorological forcings, and initial conditions (Curnel et al., 2011; de Wit and van Diepen, 2007; Hansen and Jones, 2000). However, due to the uncertainties in model structure, input data, and parameters, the model-based estimates are always subjected to errors.

To improve the model's estimates, much early effort has been directed toward the estimation of model parameters. A few classical parameter inference/inversion methods, such as batch calibration (Duan et al., 1993), pseudo-Bayesian (Freer et al., 1996), and DREAM method (Vrugt et al., 2008, 2009), have been widely accepted to minimize the residual error of modeling. To date, inverse modeling has evolved from considering parameter uncertainty alone to considering the combination of input, model structural, and parameter errors (Vrugt et al., 2005), and from considering Gaussian and stationary residual error to the inclusion of the heteroscedasticity and nonnormality of model residuals

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(Schoups and Vrugt, 2010). Among all these progress, sequential data assimilation has gained considerable popularity due to its ability of continuously updating the model states by sequentially merging new measurements. Vrugt et al. (2005) demonstrated that sequential data assimilation, together with global optimization, can provide a general framework for explicitly dealing with input, output, and model structural uncertainty by merging observations with model simulations. In the study of hydrology, observation data including soil moisture, evapotranspiration, streamflow, and water level observations have been sequentially assimilated to improve the model prediction (Crow et al., 2008; Das et al., 2008; Yu et al., 2012; Meng et al., 2017; Barthélémy et al., 2017; Zou et al., 2017). Recently, sequential data assimilation method along with crop model proved the effectiveness of improving the crop growth simulation and crop yield prediction (de Wit et al., 2012; de Wit and van Diepen, 2007; Dente et al., 2008; Huang et al., 2013; Ma et al., 2013a, 2013b; Nearing et al., 2012; Sus et al., 2014). Some studies demonstrated that even with a biased model structure, sequential data assimilation has the capability to enhance the crop model performance by updating state variables with remotely sensed observations (Ines et al., 2013). Huang et al. (2015) demonstrated that 4D variational data assimilation method is a promising way to estimate wheat yield at a regional scale, but the assimilation performance depends strongly on the leaf area index (LAI) retrieval accuracy and the scaling correction. With the rapid development of various satellites and unmanned aerial vehicles, remote sensing measurements can now provide a wide-range of soil and plant information, including surface soil moisture observations (Entekhabi et al., 1994), leaf area index (Haboudane, 2004; Verger et al., 2008), leaf nitrogen (He et al., 2016), biomass (Claverie et al., 2012), chlorophyll contents (Zhang et al., 2008), and evapotranspiration (Mutiga et al., 2010). Despite inherent uncertainties, assimilating these multi-variate observations offers a unique opportunity to better understand the intricate soil-water-atmosphere-plant system (Dorigo et al., 2007; Lakhankar et al., 2009).

The original sequential data assimilation approach has largely focused on updating state variables alone, while the updating of uncertain parameters are neglected. Nearing et al. (2012) indicated that updating-state-only often results in the discrepancies between plant state vectors and model parameters under a certain model relationship. Joint estimation of state-parameter provides a way to solve this problem. It was firstly proposed for hydrological modeling (Vrugt et al., 2005; Moradkhani et al., 2005). The parameters were assumed to be time-invariant in Vrugt et al. (2005), while dual state-parameter method by Moradkhani et al. (2005) allowed the parameters to update from one time step to the next. Joint estimation of state-parameter has proven to be a useful strategy to alleviate the inconsistency between state vectors and model parameters (Lü et al., 2011; Montzka et al., 2011; Zhang et al., 2016). Montzka et al. (2011) demonstrated that by assimilating surface soil moisture into the hydrological model via particle filter, updating both hydraulic parameters and state variables allows better prediction of top soil moisture content as compared with the result of updating-states-only. To the best of our knowledge, previous data assimilation studies for a soil-water-atmosphere-plant system focused on updating root zone soil moisture or crop yield separately. Moreover, most simultaneous estimations of states and parameters were investigated under the background of soil hydrology (Chen et al., 2015; Lü et al., 2011; Montzka et al., 2011; Shi et al., 2015) and land surface model (Andrea et al., 2017; Huang et al., 2016; Pathiraja et al., 2016; Bloom and Williams, 2015).

Joint estimation of model parameters and states for the soil-water-atmosphere-plant system is rarely studied. In this study, we will firstly develop a simultaneous state-parameter assimila-

tion strategy by combining SWAP model with ensemble Kalman filter (EnKF). Model states (e.g., soil water content, leaf area index (LAI), and grain yield) and uncertain parameters are simultaneously updated. Simultaneous state-parameter estimation (SSPE) and updating-state-only (USO) assimilation strategies are compared. The impacts of observations on soil water profile, water stress, and grain yield estimation are investigated. Although surface soil moisture (SSM) and LAI are two most easily available data sources from remote sensing measurements, we will examine the data sufficiency if multiple states and parameters are updated or if challenging situations such as stratified soils are confronted. Two basic tasks are handled by this study: (1) how well the SWAP model can work with given measurements under different circumstances, (2) If the model does not work well, what data assimilation strategy and measurement advice should be given? This study ignores the uncertainties of model structure and input data, and only considers the uncertainty of parameters in a virtual world.

In the following context, the major processes and related parameters in SWAP model are firstly described. Two assimilation methods of USO and SSPE based on EnKF are then presented. Sensitivity analysis of model parameters to model outputs based on revised Morris' method is implemented to determine the sensitive parameters. Observing simulation system (OSS) experiment is then designed to complete this study. The setup of cases and their results are discussed in Section 3. The conclusions are presented in Section 4. The objectives of this present study are: (1) with a given set of observation data, to evaluate how well the SWAP model can work and demonstrate the importance of constructing the state vector for SWAP data assimilation system; (2) to investigate the benefits and disadvantages of simultaneous state-parameter data assimilation in correcting SWAP modeling predictions, in comparison with the traditional updating-state-only data assimilation; (3) to examine the data sufficiency to simultaneously update multiple state variables and parameters under some challenging conditions and to provide measurement design advice. This investigation provides a further understanding of the response of the physically based soil-water-atmosphere-plant system to data assimilation. In return, it will help the scientists to contemplate the optimization of measurement layout for such system.

2. Method

In this section, observing simulation system (OSS) experiments are employed to assess the data assimilation performance. OSS experiments allow us to understand the interactions between observations and model predictions. Besides, OSS experiments can bypass the possible impact of model errors, which may complicate the interpretation of results. We will firstly introduce the basic principles of the forward simulation of soil water movement and crop growth and development using SWAP model. Before running data assimilation, parameter sensitivity analysis and state vector construction are carefully investigated. Two assimilation strategies of USO and SSPE based on the EnKF are then presented. A total of 13 cases is designed to investigate the assimilation performance under different conditions, including different assimilation strategies, different constructions of the state vector, different observation types, stratified soil profile and different numbers of uncertain parameters. Through these numerical experiments, we expect to provide a benchmark for developing the data assimilation of the SWAP system and to be helpful for measurement design.

2.1. SWAP model

SWAP is a one-dimensional (1-D) agrohydrological model for water flow, heat and solute transport, as well as the crop growth

and development (van Dam et al., 1997). It incorporates the WOFOST crop module, which simulates crop growth and its production accounting for water and salinity stress.

SWAP employs 1-D Richards' equation to simulate soil water flow in variably saturated soil,

$$C(h) \frac{\partial h}{\partial t} = \frac{\partial}{\partial z} \left[K(h) \left(\frac{\partial h}{\partial z} + 1 \right) \right] - S(h) \quad (1)$$

where $C(h)$ is the soil water capacity (cm^{-1}), h is the soil water pressure head (cm), z is the vertical coordinate (cm), taken positively upward, $K(h)$ is the unsaturated hydraulic conductivity function (cm/d), t is time (d), $S(h)$ is the soil water extraction rate by plant roots (d^{-1}).

Soil water retention curve is characterized by van Genuchten model (van Genuchten, 1980),

$$S_e(h) = \frac{\theta(h) - \theta_r}{\theta_s - \theta_r} = \frac{1}{(1 + |\alpha h|^n)^{1-1/n}} \quad (2)$$

and the unsaturated hydraulic conductivity $K(h)$ function is described by Mualem (1976),

$$K(h) = K_s S_e^\lambda \left[1 - (1 - S_e^{n/n-1})^{1-1/n} \right]^2 \quad (3)$$

where S_e is the degree of saturation, θ is the soil water content, θ_s and θ_r denote the saturated and residual soil water contents (cm^3/cm^3), respectively, K_s is the saturated hydraulic conductivity (cm/d), α (cm^{-1}) and n (–) are empirical shape parameters, and λ is a pore connectivity/tortuosity parameter (–), which is 0.5 in this study.

The root water uptake function $S(h)$ is the product of the root-water uptake water stress response function $\gamma(h)$ (Feddes et al., 1978) and the potential water uptake rate S_p ,

$$S(h) = \gamma(h) S_p \quad (4)$$

where $\gamma(h)$ is a prescribed function of pressure head, S_p is the product of the normalized water uptake distribution β and the potential transpiration rate T_p . In WOFOST, a crop will suffer from water stress if the soil moisture cannot meet the crop water requirement. A crop reacts to water stress with the closure of the stomata. As a consequence, reduced actual transpiration and reduced assimilation rate occur. The influence of drought on carbohydrate assimilation rate, leaf and stem death rate is quantified by introducing correcting factors due to water stress. The interaction between soil water content and crop growth is very critical for drought-sensitive crop when the soil moisture availability limits the growth. Soil water content, especially surface soil moisture, is one of the most common observations.

The WOFOST module considers two major growth driving processes (light interception and CO_2 assimilation), one growth controlling process (phenological development), and a few subsequent and accompanying processes (transpiration, respiration, partitioning of assimilates to various organs, dry matter formation). The phenological development is described by the dimensionless variable development stage (DVS), which consists of two stages for many annual crops: vegetative stage ($0 < \text{DVS} < 1$) and reproductive stage ($1 < \text{DVS} < 2$). The former corresponds to the period from seedling emergence to anthesis, and the latter is from anthesis to maturity. DVS is calculated by

$$\text{DVS}_{j+1} = \text{DVS}_j + \frac{T_{\text{eff}}}{T_{\text{sum},i}} \quad (5)$$

where j is the day number, T_{eff} is calculated from daily average temperature based on DTSMTB function (increase in temperature sum as function of daily average temperature), $T_{\text{sum},i}$ is the sum of effective temperature during the vegetative stage (named as TSUMEA

parameter) or reproductive stage (named as TSUMAM parameter). In WOFOST, many parameters, such as specific leaf area (SLATB), maximum leaf CO_2 assimilation rate (AMAXTB), reduction factor for senescence (RFSETB), fraction of total dry matter increase partitioned to roots (FRTB), fraction of above ground dry matter increase partitioned to stems, leaves, and storage organs (FSTB, FLTB, and FOTB), depend on development stage.

During the crop growth, leaf area index (LAI) is an important model state, as well as one conventional type of observation derived from remote sensing measurement. Leaf growth and senescence (caused by water stress, mutual shading, and physiologic ageing) together determine the status of leaf area index. The SPAN and TBASE are two key parameters determining the physiologic ageing. The SPAN parameter represents the maximum time in days a leaf can live at 35°C , and the TBASE parameter is defined as the lower threshold temperature for physiologic ageing. Leaf area index in WOFOST is the function of a few variables, including specific leaf area (SLATB), the dry matter weight of leaves (LV), the physiological age of leaves (LVAGE) and class number of living leaves (IVOLD).

The total dry matter in WOFOST is composed of the dry weight of roots, stems, leaves and storage organs. Photosynthesis process firstly converts CO_2 to carbohydrates (also called CO_2 assimilation), then part of the formed assimilates is assigned to the roots and the remainder is divided over the above-ground organs (stems, leaves, and storage organs). The fraction of total dry matter increase partitioned to the roots, stems, leaves and storage organs is determined by partitioning factors (FRTB, FSTB, FLTB and FOTB), which are functions of the development stage. Four parameters CVR, CVS, CVL, and CVO are used to define the conversion efficiency of assimilates into roots, stems, leaves and storage organs.

2.2. Data assimilation method

2.2.1. Ensemble Kalman filter (EnKF)

The EnKF is a variant of the Kalman filter but employs an ensemble of realizations to approximate the probability distribution function (pdf) of model states (Evensen, 1994). During the implementation of EnKF, the model parameters are firstly perturbed according to the prior information (prior mean and variance) to generate N_{ens} ensemble members. At every observation time, it assumes that all model states are linearly related to observations, and second-order pdf approximations are used to quantify the uncertainties in model states and observations. After incorporating available observations, the EnKF updates the states, then propagates the samples through time using the forward equations.

In Kalman filter system, the relation of the observation $\mathbf{y}_t$ to the model state $\mathbf{x}_t$ can be described as:

$$\mathbf{y}_t = \mathbf{H}\mathbf{x}_t + \boldsymbol{\varepsilon} \quad (6)$$

where $\mathbf{H}$ is the measurement operator that maps the model state to the observation and is an identity matrix in this study; $\boldsymbol{\varepsilon}$ is a Gaussian random error vector, $\boldsymbol{\varepsilon} \sim N[0, \mathbf{R}_t]$; $\mathbf{R}_t$ is the observation error covariance at observation time t . In this study, the potential observations are LAI, surface soil moisture, deep soil moisture, and grain yield. It is assumed that the covariance between different observations is zero, and the variance of observation will be defined in following sections.

An observation sample $\mathbf{Y}_t = [\mathbf{y}_{t,1}, \mathbf{y}_{t,2}, \dots, \mathbf{y}_{t,N_{\text{ens}}}]$ can be generated according to

$$\mathbf{y}_{t,i} \sim N[\mathbf{y}_t, \mathbf{R}_t]$$

where $\mathbf{Y}_t$ has a dimension of $N_y \times N_{\text{ens}}$, and N_y is the dimension of observation vector of $\mathbf{y}_{t,i}$ ($i = 1, 2, \dots, N_{\text{ens}}$).

The ensemble of model state predictions at time t is stored in $\mathbf{X}_t^f = [\mathbf{x}_{t,1}, \mathbf{x}_{t,2}, \dots, \mathbf{x}_{t,N_{ens}}]$ has a dimension of $N_x \times N_{ens}$ where N_x is the dimension of $\mathbf{x}_t$. The error covariance of $\mathbf{X}_t^f$ is $\mathbf{P}_t^f$. Under the linear assumptions, the estimated state $\mathbf{X}_t^a$ and related error covariance $\mathbf{P}_t^a$ is calculated as:

$$\mathbf{X}_t^a = \mathbf{X}_t^f + \mathbf{K}(\mathbf{Y}_t - \mathbf{H}\mathbf{X}_t^f) \quad (7)$$

$$\mathbf{P}_t^a = (\mathbf{I} - \mathbf{K}\mathbf{H})\mathbf{P}_t^f \quad (8)$$

where f and a are indices of the forecast and analysis estimates, respectively. $\mathbf{I}$ is the identity matrix, and $\mathbf{K}$ is the Kalman gain matrix defined as:

$$\mathbf{K} = \mathbf{P}_t^f \mathbf{H}^T (\mathbf{H} \mathbf{P}_t^f \mathbf{H}^T + \mathbf{R}_t)^{-1} \quad (9)$$

The error covariance between forecast and analysis is directly calculated from an ensemble of model simulations:

$$\mathbf{P}_t^f \mathbf{H}^T = (N_{ens} - 1)^{-1} \sum_{n=1}^{N_{ens}} (\mathbf{x}_{t,n}^f - \bar{\mathbf{x}}_t^f) (\mathbf{H} \mathbf{x}_{t,n}^f - \mathbf{H} \bar{\mathbf{x}}_t^f)^T \quad (10)$$

where $\bar{\mathbf{x}}_t^f$ represents the ensemble mean calculated as:

$$\bar{\mathbf{x}}_t^f = \frac{\sum_{n=1}^{N_{ens}} \mathbf{x}_{t,n}^f}{N_{ens}} \quad (11)$$

2.2.2. Updating state only and simultaneous state-parameter estimation assimilation strategies

Two different assimilation strategies, i.e., USO and SSPE, are introduced to update the state vector based on the EnKF. USO strategy does not update the uncertain parameters, but only updates the model states at the observation time. USO assimilation strategy has been widely used in estimating model states. However, Nearing et al. (2012) demonstrated that only updating state often results in plant state vectors which disagree with model parameters. Similarly, Curnel et al. (2011) found that the uncertainty of TSUMEA and crop emergence date may cause the “phenological shift” phenomenon and resulting poor data assimilation performance. In order to alleviate the inconsistency between parameters and updated states, three types of algorithms are typically used for simultaneous estimation of states and parameters, including state vector augmentation approach (Monsivais-Huertero et al., 2010; Zhang et al., 2016), recursive dual filter approach (Moradkhani et al., 2005), parameter optimization and state assimilation approach (Vrugt et al., 2005). In this study, the augmented state vector is adopted to carry out the SSPE strategy due to the simplicity of this algorithm.

In state augmentation, the state vector $\mathbf{x}_{t,i}$ of i th ensemble at time t and the measurement operator $\mathbf{H}$ consist of two blocks:

$$\mathbf{x}_{t,i} = (\mathbf{s}_{t,i}, \mathbf{p}_{t,i}) \quad (12)$$

$$\mathbf{H}^* = (\mathbf{H}_s, \mathbf{H}_p) \quad (13)$$

where $\mathbf{s}$ refers to model states and $\mathbf{p}$ to parameters. The error covariance between forecast and analysis and Kalman gain is now given by:

$$\mathbf{P}_t^{sf} \mathbf{H}^{*T} = (N_{ens} - 1)^{-1} \sum_{n=1}^{N_{ens}} (\mathbf{x}_{t,n}^f - \bar{\mathbf{x}}_t^f) (\mathbf{H}^* \mathbf{x}_{t,n}^f - \mathbf{H}^* \bar{\mathbf{x}}_t^f)^T \quad (14)$$

$$\mathbf{K}^* = \mathbf{P}_t^{sf} \mathbf{H}^{*T} (\mathbf{H}^* \mathbf{P}_t^{sf} \mathbf{H}^{*T} + \mathbf{R}_t)^{-1} \quad (15)$$

The estimated state $\mathbf{X}_t^a$ and related error covariance $\mathbf{P}_t^a$ is then calculated as:

$$\mathbf{X}_t^a = \mathbf{X}_t^f + \mathbf{K}^*(\mathbf{Y}_t - \mathbf{H}^* \mathbf{X}_t^f) \quad (16)$$

$$\mathbf{P}_t^a = (\mathbf{I} - \mathbf{K}^* \mathbf{H}^*) \mathbf{P}_t^{sf} \quad (17)$$

2.3. Observing simulation system (OSS) experiment

OSS experiment is a virtual numerical study by using synthetically generated data instead of in-situ observation. OSS experiment consists of $N_{ens} + 1$ members sampled from a modeling uncertainty pdf. One of these simulations is selected as the truth system, and the remaining N_{ens} members are used to run data assimilation and the open-loop simulation (without data assimilation). The truth system is used to generate synthetic observations and evaluate the data assimilation performance as well. An ensemble size of 500 is chosen.

2.3.1. Data and model set-up

Meteorological data are taken from the Wageningen weather station (<http://www.swap.alterra.nl/>), including solar radiation, minimum and maximum air temperature, relative humidity, wind speed and precipitation at a daily frequency from 1st May to 15th October 1981 (totally 168 days). In all simulations, the upper boundary condition is atmospheric boundary condition. The potential evapotranspiration (ET_p , cm/d) is estimated by the Penman-Monteith equation (Monteith, 1981). The ET_p is then partitioned into potential soil evaporation (E_p , cm/d) and potential crop transpiration (T_p , cm/d). The actual transpiration (T_a , cm/d) is governed by root water uptake which is calculated from the root length density distribution, potential transpiration and water stress (Bouten, 1992; Skaggs et al., 2006). The actual evaporation equals its potential rate E_p in wet soil but decreases into the minimum value of E_p and E_{max} (calculated by Darcy's law) in dry soil. In addition, the bottom boundary condition is defined as a free drainage situation, and the bottom flux depends directly on the hydraulic conductivity of the lowest compartment. In this study, the crop is maize, and the maximum root depth is set as 100 cm with an initial root depth of 10 cm at emergence and a potential daily increase rate of 1.5 cm d⁻¹.

2.3.2. Parameter sensitivity analysis

In this section, parameter sensitivity analysis is implemented to identify the sensitive parameters for several key model output. Fifty-six model parameters related to soil water flow, plant development, light-use efficiency, dry matter partitioning and accumulation, and the growth of leaf, root, stem and storage organ are selected (Appendix A). Sensitivity analysis of these 56 parameters to leaf area index (LAI), surface soil moisture (SSM), development stage (DVS), water stress (WS) and final grain yield is addressed. The mean values of parameters used for sensitivity analysis are collected from the references and documentation (Boons-Prins et al., 1993; Carsel and Parrish, 1988; Wang et al., 2013; Ceglar et al., 2011), and the variation range is set as $\pm 10\%$. All parameter distributions are assumed to be normal. The reference bound of each parameter is given in Table 4.

Morris' method (1991) is employed to determine the few most influential parameters among a large number of parameters. In Morris, each parameter x_i is assumed to vary over p selected levels. For a given value of $X = (x_1, x_2, \dots, x_n)$, the elementary effect of parameter is calculated as

$$R_i(x_1, \dots, x_n, \Delta) = \frac{y(x_1, \dots, x_{i-1}, x_i + \Delta, x_{i+1}, \dots, x_n) - y(x_1, \dots, x_n)}{\Delta} \quad (18)$$

where Δ is a multiple of $1/(p - 1)$, p is the number of levels, $x_i + \Delta$ is the transformed point where only the component i has been

changed. The total number of model evaluations needed is $r \cdot (n + 1)$. r is the number of trajectories ($r = 500$), and the value of parameters in each trajectory is generated in normal distribution based on the average and standard deviation in Table 4.

There are two sensitivity measures: μ and σ (Morris, 1991). The first measure μ describes the overall influence of the parameter on $y(X)$, and the second measure σ estimates the totality of the higher order effects of all the R_i obtained for each parameter. Here, a revised version of the measure μ , named as μ^* , is used to rank the parameters in order of importance (for details see Campolongo et al. (2004)). μ^* is an estimate of the mean distribution of the absolute values of the elementary effects R_i . It has demonstrated that the use of measure μ^* as a proxy of variance-

based total index is convenient. Morris' method and its revised version have been widely used due to its simplicity and validity (for example, Ge and Menendez, 2017; Gan et al., 2014; King and Perera, 2013). In this study, the calculated sensitivity index μ^* with respect to model outputs is presented in Fig. 1.

2.3.3. Construction of state vector

An appropriate construction of updated state vector is critical to implement data assimilation. Soil moisture is one fundamental component of state vector commonly updated by many studies since it acts as the primary control on crop water supply (de Wit and van Diepen, 2007). However, the selection of state vector for crop-growth related variables in the literature is considerably

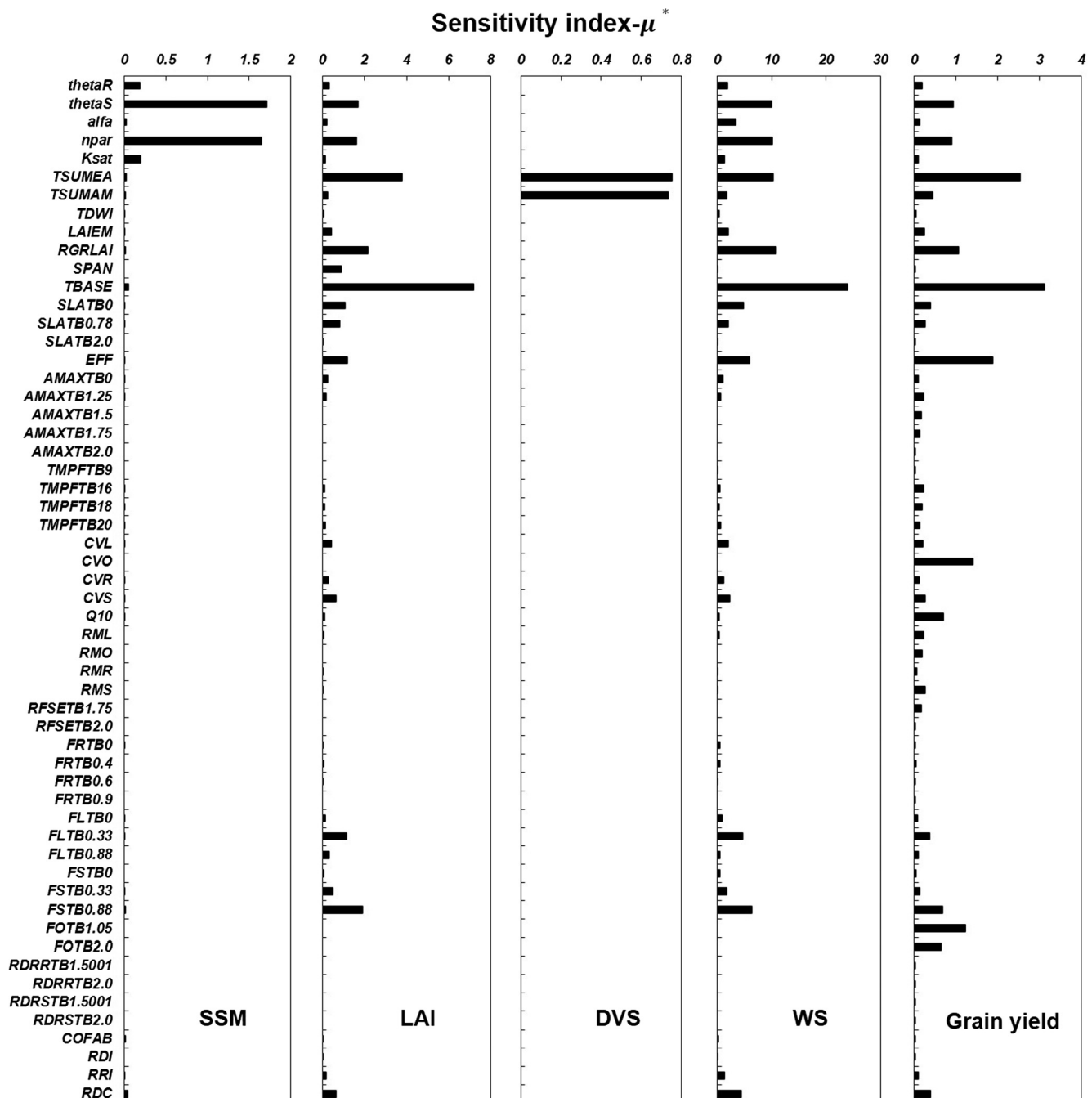


Fig. 1. The sensitivity index of parameters to surface soil moisture (SSM), leaf area index (LAI), development stage (DVS), water stress (WS), and final grain yield. The definition of each parameter refers to Appendix A.

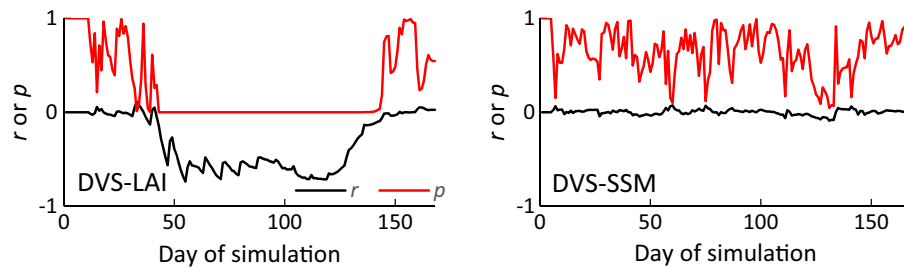


Fig. 2. The temporal evolution of correlation between LAI (SSM) and DVS and the confidence level for Case 3. r -correlation coefficient; p -confidence level (from Pearson's correlation method, significant confidence level $p < .01$).

Table 1

State vector candidates (variable component).

Potential component	SV 1	SV 2	SV 3	SV 4
Soil moisture	✓	✓	✓	✓
Specific leaf area		✓	✓	✓
Leaf structural dry matter		✓	✓	✓
Leaf age		✓	✓	✓
Class number of the oldest leaf		✓	✓	✓
LAI	✓			
Development stage			✓	✓
Grain yield				✓

diverse. Vazifedoust et al. (2009) selected LAI and relative transpiration as the internal variables to be updated, and crop dry matter and dry matter of storage organ are sequentially updated based on the ratio of updated and simulated LAI. Curnel et al. (2011) and Huang et al. (2015) chose LAI as the unique state to implement data assimilation. Some other researchers did not update LAI directly since LAI value is a function of several variables. In the data assimilation study of DSSAT crop model, LAI is indirectly updated by updating plant leaf area and plant leaf weight with LAI data (Ines et al., 2013; Nearing et al., 2012).

If all the model variables of SWAP are included in the state vector, spurious updates may be caused when state vector contains components that are not well related to the components of observation vector. A preliminary correlation analysis between model variables and observations is thus suggested, so that local approximately linear correlated variables to observations can be selected (Nearing et al., 2012).

In this study, LAI and SSM are two major observations. The correlations between model variables and observations are firstly calculated to preliminarily choose the variables to be updated. An example of the temporal evolution of correlation between DVS and LAI or SSM over simulation period will be presented in Fig. 2. Four different choices of state vectors (SV) are investigated to reveal the consequence of each choice. Note that this paper focuses on soil moisture, water stress, leaf area index, and grain yield estimation, thus variables related to root and stem growth are neglected.

SV1: Both soil moisture and LAI are directly updated. LAI is both an output variable and an important intermediate variable in SWAP system, and many critical processes (for example, transpiration and dry matter accumulation) are strongly influenced by LAI. If updating SV1 can lead to satisfactory estimates to these critical processes, the correlation analysis will be not necessary. Most existing studies directly update LAI.

SV2: LAI is not updated directly, instead its four dependent variables, including the specific leaf area (SLA), the dry matter weight of leaves (LV), the physiological age of leaves (LVAGE) and class number of living leaves (IVOLD), are updated. Correlation analysis has shown that all these four variables are highly correlated with LAI (figure not shown).

SV3: Development stage (DVS) is further included based on SV2. The correlation between LAI and DVS shows obvious temporal variability, only manifesting strong correlation at the middle stage of simulation period (Fig. 2). Most existing data assimilation studies did not update DVS. Since many parameters and variables in crop model depend on DVS, we will show the consequence of updating DVS.

SV4: Grain yield (GY) is further included based on SV3. GY and LAI show large correlation at the early reproductive stage (figure not shown). However, for a physically based model, it is intuitive that grain yield can be automatically renewed after the driving variables (such as soil moisture and LAI) and parameters have been updated. It is not clear if a further direct correction to grain yield is beneficial or not. Comparison between SV3 and SV4 is used to examine this.

Table 1 lists the candidate state vectors. The state vectors in Table 1 only consist of model variables. Parameter component will be augmented when SSPE assimilation strategy is employed. Here, it is noted that although a correlation assessment can be run to pre-select the state variables highly correlated with observations, EnKF cannot guarantee mutually consistent model states after each update due to the inherent model nonlinearity in SWAP system and because EnKF uses a single correlation relationship to update ensemble member regardless of the current growth stage of each particular simulation (Nearing et al., 2012). Furthermore, the cross-correlation between variables and observations may also change under different conditions (such as under varying levels of drought stress).

2.3.4. General description of synthetic experiments

A general description of numerical experiments to serve the objectives of this study is presented. A total of 13 cases are designed. The true and prior information of considered uncertain parameters for all cases is given in Table 2. Considering there are many cases serving relevant purposes, the detailed setup of each case and corresponding results will be presented separately in each subsection of Section 3.

Only one layer of loam soil with 200 cm depth is considered in Cases 1–7, in which seven uncertain parameters (component P1 in Table 2) are selected by the sensitivity analysis results. These seven cases are used to investigate how well the SWAP model (with unknown parameters) can perform under given observations of LAI and SSM but with different state vector constructions and different data assimilation strategies. Cases 1–4 are to examine the influence of state vector construction on data assimilation performance. USO strategy is employed for these four cases, and each state vector has been defined in Table 1. SSPE strategy is employed in Cases 5–7. Comparison between Case 4 and Case 5 is to evaluate the different outcomes of SSPE and USO strategies. Comparison among Cases 5–7 is to reveal the outcomes by assimilating LAI and SSM together, assimilating LAI alone, and assimilating SSM alone. A two-layer soil profile is considered in Cases 8–10: the

Table 2Unknown parameters for each case. s.d.-standard error. The definitions of parameters can be found in [Appendix A](#).

Component			Parameter	Unit	True value	Prior value	Uncertainty (s.d.)	
P3 (Case 11-13)	P2 (Case8-10)	P1 (Case 1-7)	thetaS1	cm ³ cm ⁻³	0.43	0.4	0.04	
			Npar1	-	1.56	2	0.2	
			Ksat1	cm d ⁻¹	24.96	50	15	
			TSUMEA	°C d	695	900	180	
			RGR LAI	m ⁻² m ⁻² d ⁻¹	0.0294	0.035	0.007	
			TBASE	°C	10	7	1.5	
			CVL	kg kg ⁻¹	0.68	0.75	0.075	
				</				

upper layer of loam is from 0 to 30 cm depth, and the lower layer of silt loam is from 30 to 200 cm depth. Comparing to Cases 1–7, three more uncertain soil parameters are introduced in Cases 8–10 due to the soil stratification (component P2 in [Table 2](#)). Cases 8–10 are utilized to investigate how well the SWAP model (with more unknown soil parameters) can perform under given observations of LAI and SSM in the stratified soil. Measurement design advice is accordingly given.

Based on Case 10, Cases 11–13 introduce two more uncertain crop parameters EFF and CVO (component P3 in [Table 2](#)). Both EFF and CVO have high correlations with grain yield (GY), however, CVO is not sensitive to LAI and SSM. Comparison between Cases 11–13 is used to examine the necessity of collecting more types of crop data if more crop parameters are poorly known.

3. Results and discussion

3.1. Influence of state vector construction

Cases 1–4 are to discuss the influence of state vector construction on assimilation performance. The motivation is driven by the fact that so many different state vectors have been constructed in the literature for SWAP system, it is not clear how well the SWAP model (with poorly known parameters) will perform under the continuous integration of LAI and SSM observations. Only USO strategy is used in this subsection. The state vectors of Cases 1–4 are SV1, SV2, SV3 and SV4 (defined in [Table 1](#)), respectively. Case 1 is designed to test the appropriateness of the most common state vector used in the literature. In Case 2, we will demonstrate the

necessity of updating SLA, LV, LVAGE, and IVOLD when WOFOST model is used. Case 3 aims to reveal the importance of updating development stage (DVS), and the cause of “phenological shift” is postulated. “Phenological shift” is a phenomenon that all the members of the ensemble are not at the same phenological stage, which is caused by the uncertainty on emergence date and TSUMEA parameter ([Curnel et al., 2011](#)). Case 4 will examine the necessity of introducing grain yield as the component of the state vector. A summary of Cases 1–4 is presented in [Table 3](#).

The observation data of LAI and SSM (0–5 cm depth average) in Cases 1–4 is synthesized by running the SWAP model with true parameter values. The observation intervals of SSM and LAI are 3-day (3, 6, 9, ..., 165 day of simulation) and 8-day (8, 16, 24, ...,

Table 3

Cases summary.

Case	Soil layer number	Observation type	Assimilation strategy	State vector
1	1	LAI + SSM	USO	SV1
2	1	LAI + SSM	USO	SV2
3	1	LAI + SSM	USO	SV3
4	1	LAI + SSM	USO	SV4
5	1	LAI + SSM	DSPE	SV4 + P1
6	1	LAI	DSPE	SV4 + P1
7	1	SSM	DSPE	SV4 + P1
8	2	LAI + SSM	DSPE	SV4 + P2
9	2	LAI(0.1)+SSM(0.01)	DSPE	SV4 + P2
10	2	LAI + SSM+SM40	DSPE	SV4 + P2
11	2	LAI + SSM+SM40	DSPE	SV4 + P3
12	2	LAI + SSM+SM40	USO	SV4
13	2	LAI + SSM+SM40 + GY	DSPE	SV4 + P3

160 day of simulation), respectively. The uncertainty of SSM observation is constant with an observation error variance of $0.04^2 \text{ m}^3 \text{ m}^{-3}$. The uncertainty of LAI observation is time-dependent. The low uncertainty of LAI observation is set at the early stage, then the uncertainty increases as the crop grows. The observation error variance of LAI observation at $t = 8, 16, 24, 32, 40, 48, 56, 64, 72, 80$ day of simulation period are $0.1^2, 0.1^2, 0.15^2, 0.2^2, 0.2^2, 0.25^2, 0.3^2,$

$0.3^2, 0.35^2, 0.4^2 \text{ m}^2 \text{ m}^{-2}$, respectively. After 80 days of simulation, the observation error variance of LAI observation is retained at $0.4^2 \text{ m}^2 \text{ m}^{-2}$. LAI and SSM observations are jointly assimilated in these four cases.

The mean estimates and associated uncertainties of Cases 1–4 are presented in Fig. 3. The uncertainty is evaluated by spread, which is given by,

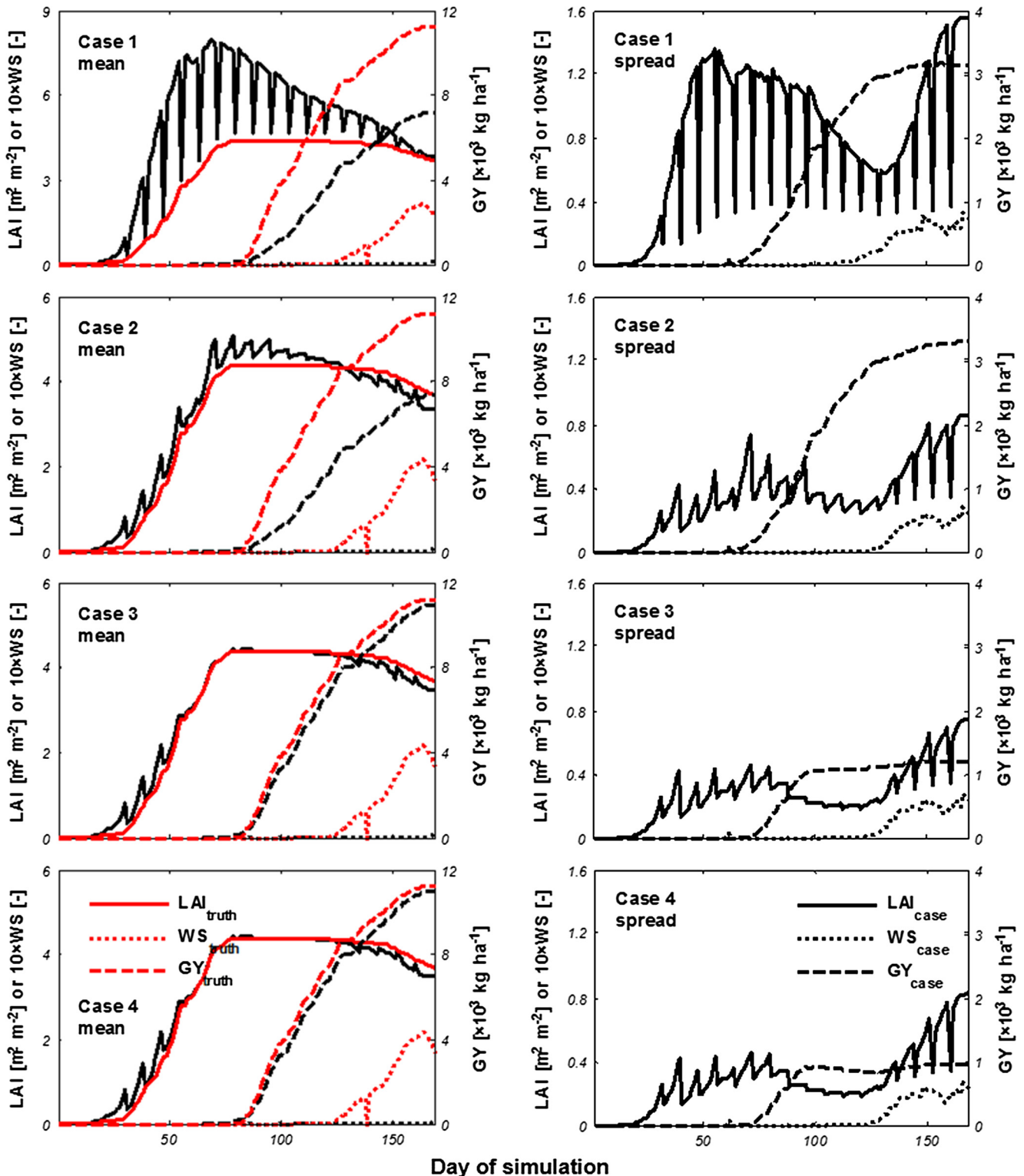


Fig. 3. The temporal evolution of LAI, water stress (WS) and grain yield (GY) over the simulation period. The red lines represent the results of the true run; the black lines represent the results of Case 1–4.

$$\text{spread} = \sqrt{\frac{\sum_{n=1}^{N_{\text{ens}}} (Y_n - \bar{Y})^2}{N_{\text{ens}} - 1}} \quad (19)$$

$$\bar{Y} = \frac{\sum_{n=1}^{N_{\text{ens}}} Y_n}{N_{\text{ens}}} \quad (20)$$

where Y_n represents the simulated LAI, water stress, or grain yield of ensemble n . The smaller the spread, the less uncertainty with the estimated results.

As shown in Fig. 3, by using SV1, LAI only shows corrections at the observation time, and it bounces back quickly to a deviated value in comparison to the true LAI. By updating four dependent variables of LAI (SV2), the estimation of LAI development is largely improved. Comparing to Case 1, a smaller spread of LAI is also observed for Case 2. The inferior performance of using SV1 indicates that a direct update of LAI is not appropriate when sequential data assimilation method is employed in SWAP model, such that recalibration-based assimilation approach may be used to acquire satisfactory performance (Curnel et al., 2011; de Wit et al., 2012; Ma et al., 2013a, 2013b). The underlying reason for the failure of SV1 is that low LAI observation frequency (once every 8 days) cannot assure the correct evolution of LAI. Case 2 demonstrates the necessity of simultaneous updating of specific leaf area, the dry matter weight of leaves, the physiological age of leaves and class number of living leaves. Although results are not presented here, we found that solely updating specific leaf area and the dry matter weight of leaves but ignoring updating physiological age of leaves and class number of living leaves (the state vector used by Nearing et al. (2012) with DSSAT model) is not adequate if SWAP model is used.

It is clear that neither Case 1 nor Case 2 can produce a satisfactory estimation of grain yield (Fig. 3). However, in Case 3 by further updating development stage, a clear improvement to LAI and grain yield estimation is obtained. The final grain yield of Case 3 is 10893 kg ha^{-1} , which is very close to the observed final grain yield (11241 kg ha^{-1}). The results demonstrate that grain yield can be readily renewed after some driving components (such as LAI and DVS) being updated. Fig. 4 compares the estimated DVS in Case 2 and Case 3. It is shown that the estimation of DVS in Case 3 is significantly improved after 50 days. From Fig. 2, firstly, we can infer that SSM observation is of no use to update DVS due to the low correlation between them over the entire simulation period. Secondly, LAI and DVS have a weak correlation at the early time because of the small total dry matter increase and limited influence of DVS on LAI development. As the crop grows, the correlation between LAI and DVS increases since parameters determining LAI development becomes more dependent on DVS value. The benefit of updating DVS is further demonstrated by Fig. 5a and b, which plot

the temporal variation of DVS ensembles, as well as the mean and true DVS from 1st to 80th day. The FLTB function values versus DVS are also given. In Fig. 5a and b, the FLTB function is composed of three linear segments (when $0 < \text{DVS} < 0.33$, $\text{FLTB} = 0.7$; when $0.33 < \text{DVS} < 0.88$, FLTB linearly decrease from 0.70 to 0.15; when $0.88 < \text{DVS} < 0.95$, FLTB linearly decrease from 0.15 to 0). The ensembles of DVS for Case 2 have a wide variation range, and the estimated mean DVS fails to match the true DVS, but the estimated DVS for Case 3 agrees well with the true values especially after 50 days. Comparing Fig. 5c and Fig. 5d, the more accurate DVS estimation in Case 3 leads to a better estimation of FLTB value and subsequently better LAI estimation. In the work of Curnel et al. (2011), it was found that all the members of the ensemble was not at the same phenological stage (caused by the uncertainty on emergence date and TSUMEA parameters in their study), and the so-called “phenological shift” phenomenon greatly damaged the performance of EnKF-based assimilation method. A recalibration-based assimilation approach for WOFOST crop model was thus suggested by Curnel et al. (2011). However, according to our results, a direct updating of DVS renders EnKF-based strategy effective.

By comparing the results of Case 3 and Case 4 (Fig. 3), the spread of the grain yield in Case 4 is slightly lower than that in Case 3, while mean grain yield estimations do not have obvious differences between two cases (final yield: Case 3: 10890 kg ha^{-1} , Case 4: 11010 kg ha^{-1} , observation: 11241 kg ha^{-1}). Although in this particular case study, further updating grain yield is not significantly beneficial, we recall that there is a strong correlation between LAI and grain yield during the early time of reproductive stage (figure not shown). In another case study, we find that if DVS is not updated, then updating grain yield is obviously helpful to correct the grain yield estimation (figure not shown).

In summary, the construction of state vector strongly influences the data assimilation performance. The specific leaf area, the dry matter weight of leaves, the physiological age of leaves, the class number of living leaves, and development stage should be included in the state vector.

3.2. Comparison of USO and SSPE assimilation strategies

Most existing data assimilation study for soil-water-atmosphere-plant system ignored updating parameters, mainly because of the large number of parameters in such system. The effect of updating parameter in a sequential way could be two-sided, especially in a complex system with a large number of unknowns (Song et al., 2014). The benefit is obvious since correct parameter estimates can enhance the model predictability, while the potential damage is also notorious, for example, parameter updates may partially compensate for model structural error and

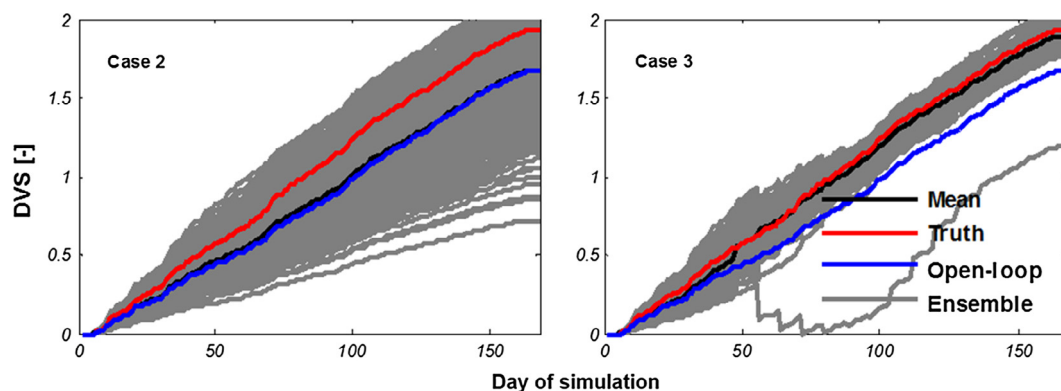


Fig. 4. The temporal evolution of estimated developmental state (DVS) of Case 2 and Case 3.

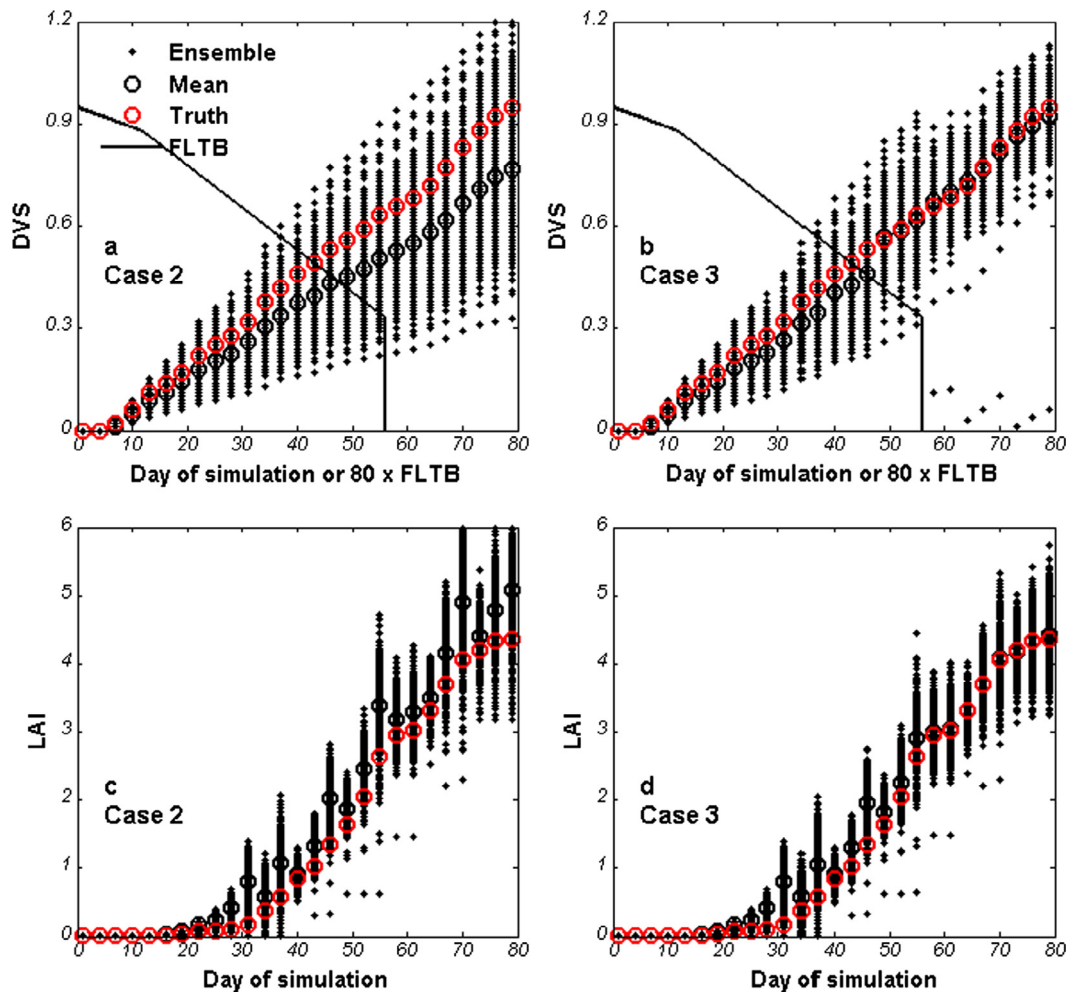


Fig. 5. The estimated DVS versus time (and FLTB function) and estimated LAI from 1st to 80th day. FLTB: fraction of total above ground dry matter increase partitioned to the leaves as a function of DVS.

input error (Schoups and Vrugt, 2010). It is also sensible that limited information (such as LAI and SSM) may impose poor constraint on parameter estimation and consequently cause biased estimation, similar to the nonuniqueness issue in an ill-posed inverse problem (Keating et al., 2010). For the soil-water-atmosphere-plant system, it is thus necessary to understand how well the SWAP model can perform by updating the parameters with given observations (observations are usually very limited in the practical application of soil-water-atmosphere-plant problem). Based on Case 4, Case 5 is designed by augmenting the parameters (component P1 in Table 2) with the state variables. The observation data of LAI and SSM is synthesized by running the SWAP model with reference parameter values. The uncertainty of observation is set to be same with that in Case 4. LAI and SSM observations are simultaneously assimilated in Case 5. Comparison between Case 4 and 5 is expected to demonstrate different outcomes of USO and SSPE strategy. LAI and SSM observations are also separately assimilated in Case 6 and 7. Comparison among Case 5–7 can further reveal the impact of data availability on parameter estimation for the soil-water-atmosphere-plant system. A similar comparison has been studied by Ines et al. (2013) but without updating of model parameters. A summary of Cases 5–7 is presented in Table 3.

Fig. 6 presents the temporal evolution of mean soil water profile and associated spread of the open-loop case and Cases 4–5. Compared to the open-loop simulation, the estimated soil water profile in Case 4 received substantial improvement, while there is still a

large gap compared with the true value. Fortunately, by integrating the parameters in the state vector (Case 5), the temporal change of soil water profile is perfectly reproduced with much smaller uncertainty. SSPE strategy also produces much better estimates for LAI and grain yield (Fig. 7). The comparison between Case 4 and Case 5 clearly demonstrates the benefits of SSPE strategy in terms of water stress, LAI and grain yield estimation.

It is expected that the superior outcomes of Case 5 are due to the updating of model parameters. Fig. 8 shows the estimated parameters from Cases 5, 6 and 7 by using different types of data. From the results of case 6, we can find that assimilating LAI observation leads to improved estimates for all uncertain parameters except Ksat1. The accuracy of crop parameter estimation (TSUMEA, TBASE, RGRLAI, and CVL) is fairly good. It is interesting to see that continuously assimilating LAI data can provide some corrections to soil parameter ThetaS1 and npar1, and the improvement mainly happens at the late time of simulation period. This is because that water stress starts to affect the LAI development, and water stress level is influenced by the soil parameters. The parameter estimation performance of Case 6 can be confirmed by the correlation analysis results (not presented), which show that LAI is correlated with the other six unknown parameters but Ksat1. From the results of Case 7, SSM data only derives an agreeable estimate for soil parameter npar1 while estimated Ksat1 and ThetaS1 fail to converge to their true values. The soil parameter estimation performance is also consistent with the correlation analysis. The poor

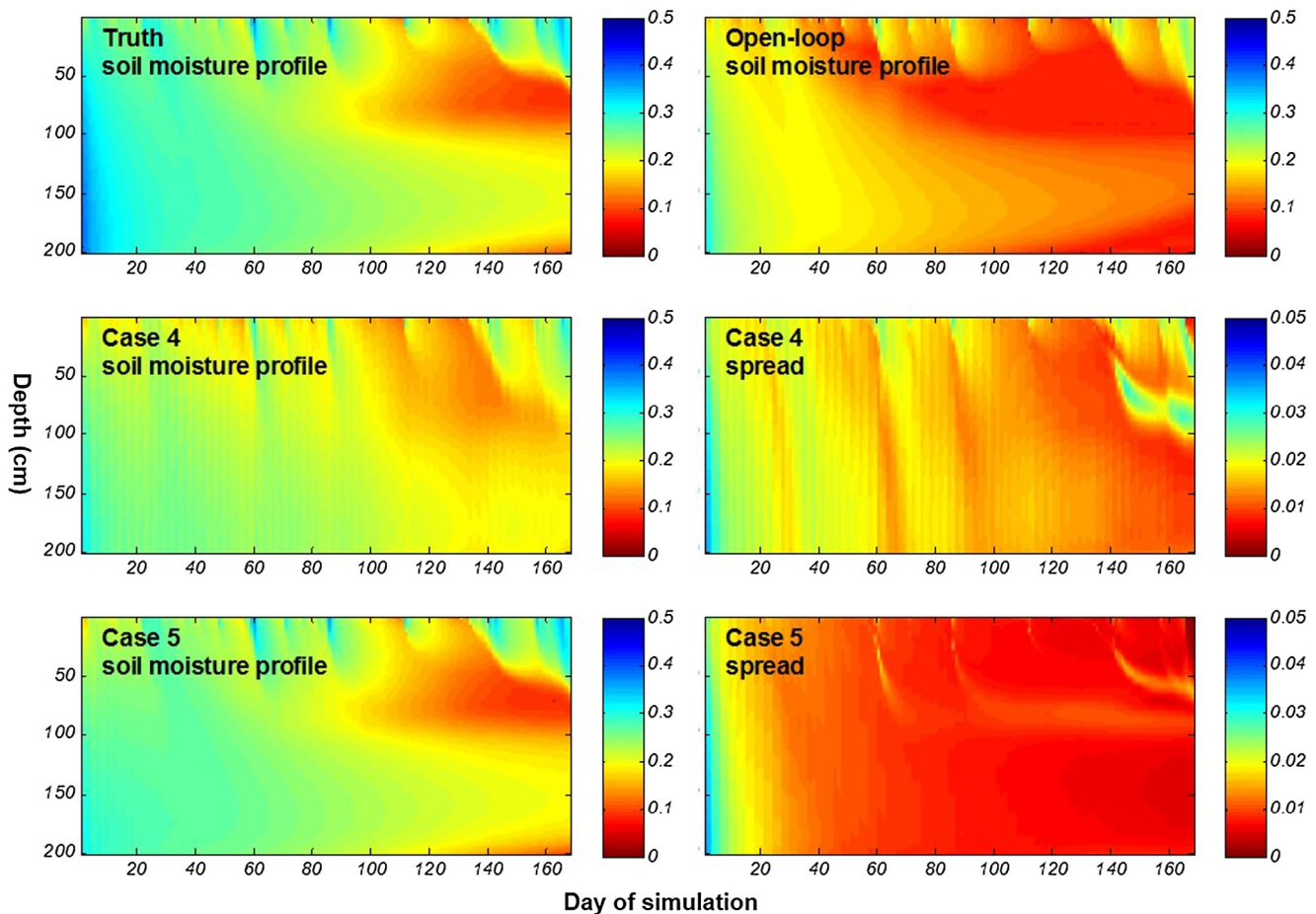


Fig. 6. The temporal evolution of mean soil water content profile and the spread in case 4 and 5 over the entire simulation period.

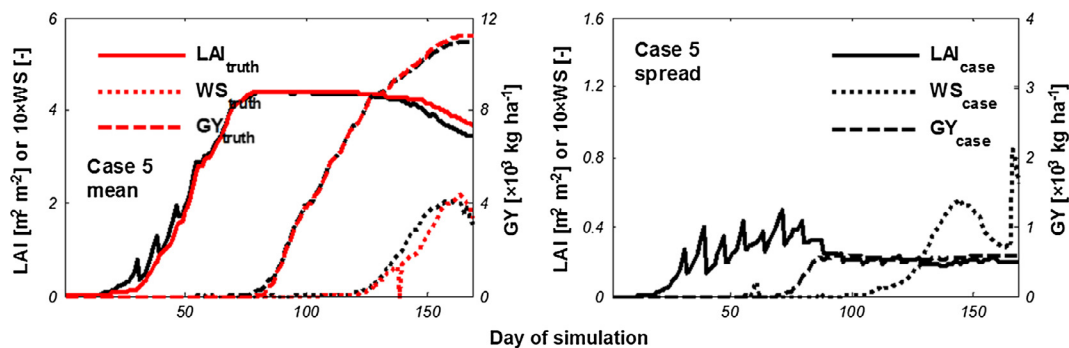


Fig. 7. The temporal evolution of mean LAI, WS, and GY and associated spread in case 5 over the entire simulation period.

performance of estimating Ksat1 and ThetaS1 is understandable since the soil profile in this study does not experience saturation (Fig. 6) under given rainfall intensity. In another numerical experiment with heavier rainfall, it is found Ksat1 and ThetaS1 can be perfectly estimated if the soil suffers drainage (results not shown here). Unlike the positive effect of LAI observation on soil parameter estimation, SSM observation is worthless in estimating crop parameters due to the low correlation between SSM and crop parameters in this study. Comparing Cases 6 and 7 with Case 5, it indicates that simultaneous assimilation of LAI and SSM in Case 5 brings trivial benefit comparing to only using either LAI or SSM. In reality, the practical role of data on parameter estimation is also

influenced by observation frequency and error, which is not discussed in this paper.

Although LAI (or SSM) averagely provides limited worth in soil (or crop) parameter estimation, simultaneously assimilating LAI and SSM is helpful to assure a successful analysis (Figs. 6 and 7). However, the temporal evolutions of estimated soil water profile, LAI, grain yield, and water stress from Case 6 and Case 7 show big differences by separately assimilating LAI and SSM observations (Fig. 9). Recalling the true soil moisture profile in Fig. 6, it is seen that assimilating LAI alone (Case 6) does not have the ability to reproduce the soil moisture well. Similarly, assimilating SSM alone (Case 7) is not able to generate satisfactory LAI, grain yield,

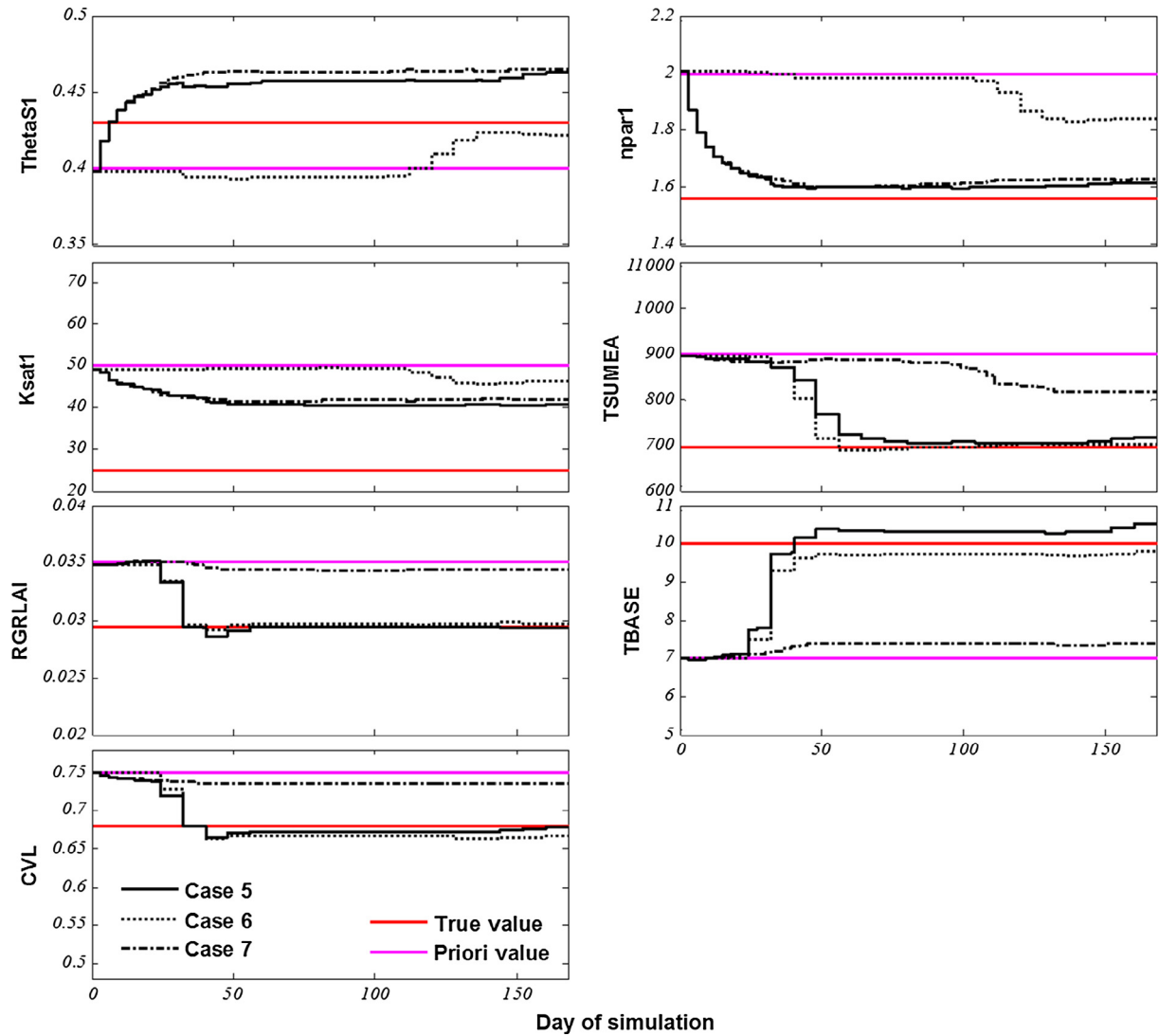


Fig. 8. The temporal evolution of parameter estimates of Cases 5–7 over the simulation period.

and water stress. It indicates that assimilating two types of data together always outperforms only using one type of observation. In Case 5, all estimates are better than these in Cases 6 and 7.

Another advantage of SSPE assimilation strategy is that this strategy has alleviated the inconsistency between model states and model parameters by updating uncertain parameters. Following the definition in (Song et al., 2014), the inconsistency is measured via

$$IC_t = |S_{a,t} - S_{m,t}| \quad (21)$$

$$S_{m,t} = M(v_0, u_t) \quad (22)$$

where $S_{a,t}$ is the estimated model states at observation time t ; v_0 is the initial state variables at time zero; u_t is the model parameter set. In case 5, u_t includes the updated model parameters, while, in case 4, u_t includes the priori value of model parameters. $S_{m,t}$ is the model states by rerunning SWAP model from time zero. Smaller IC_t values suggest lower inconsistency between model states and parameters.

The inconsistency between model states of LAI, DVS, grain yield, soil moisture, and model parameters are calculated for Cases 4 and 5. The SSPE assimilation strategy has obviously alleviated the inconsistency between model states and model parameters (Fig. 10).

From the analysis above, a SSPE assimilation strategy is more useful than a USO strategy to obtain better soil water and crop growth estimation. The parameter estimation accuracy is influenced by the data type and the correlation between observation and parameter. Crop parameters are relatively easier to estimate by infusing LAI observations. However, estimating soil parameter is challenging, which we attribute to the strong nonlinearity of the soil water equation and limited parameter information carried by surface soil moisture data. The degree of interaction between soil water and plant may also affect the parameter estimation. For example, it will be difficult to infer soil parameters using only LAI data if water stress does not exist. An additional benefit of the SSPE method is the reduction of inconsistency between variables and parameters.

3.3. Stratified soil profile

The stratified soil profile is common in real-world cropland. Cases 8–10 are implemented to investigate the performance of SSPE assimilation strategy in the stratified soil profile. A two-layer soil profile is considered: the upper layer of loam is from 0 to 30 cm depth, and the lower layer of silt loam is from 30 to 200 cm depth in Cases 8–10. There are totally ten uncertain parameters, including three soil parameters for each layer and four

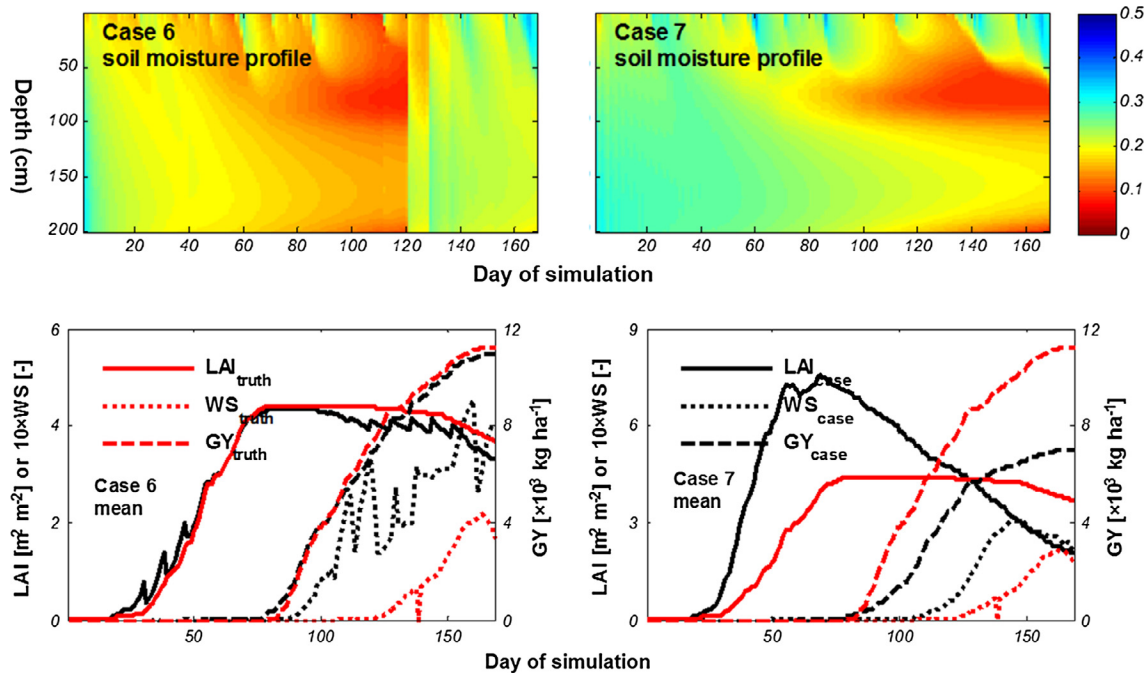


Fig. 9. The temporal evolution of soil moisture profile and crop growth outputs of case 6 and case 7 over the simulation period.

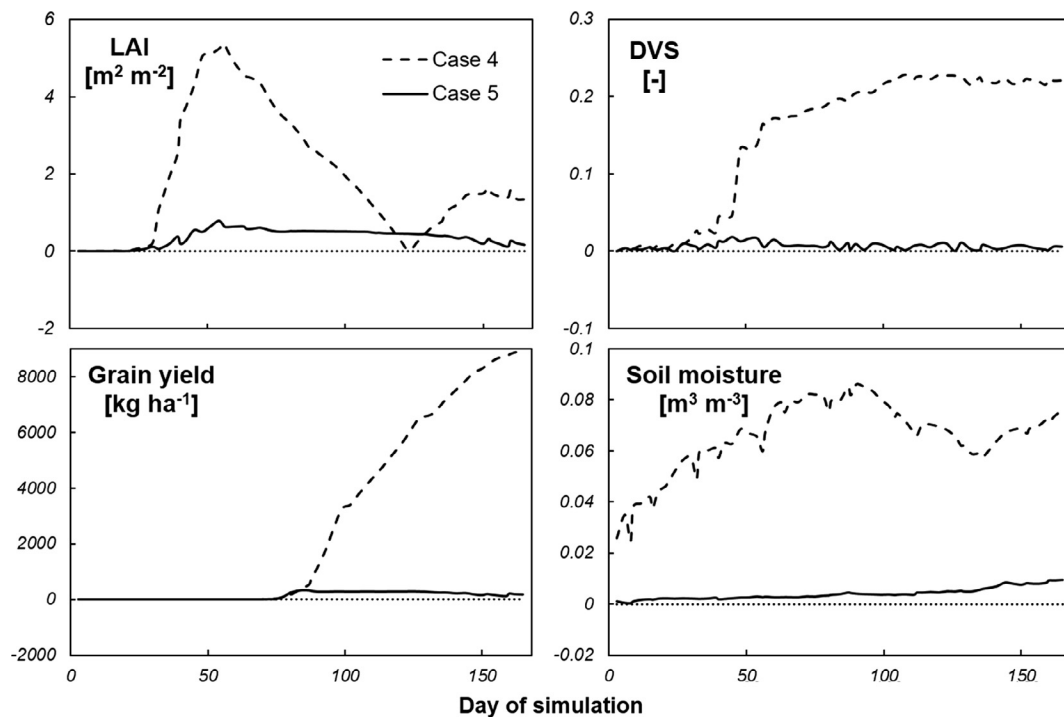


Fig. 10. The temporal evolution of inconsistency between model states and model parameters.

crop parameters (component P2 in Table 2). In Cases 8–10, three more uncertain soil parameters are introduced due to the stratified soil profile comparing to Case 5. LAI and SSM observations synthesized by using true parameter values are simultaneously assimilated in Case 8. The observation intervals and uncertainties of LAI and SSM observations are same with that in Case 1–7. To test the influence of observation error on the parameter and state estimation, comparing to Case 8, the observation error variances of LAI and SSM in Case 9 are reduced into constant values of 0.1^2 m^2

m^{-2} and $0.01^2 \text{ m}^3 \text{ m}^{-3}$, respectively. Based on Case 8, to investigate whether another observation is needed for improving assimilation performance under stratified soil profile condition, Case 10 incorporates one more point-scale soil moisture observation at 40 cm depth (name SM40 with constant error variance of $0.04^2 \text{ m}^3 \text{ m}^{-3}$) with observation interval of 3-day. A summary of Cases 8–10 is presented in Table 3.

Parameter estimation is firstly discussed. The temporal evolution of estimated parameters varies across Cases 8–10 over the

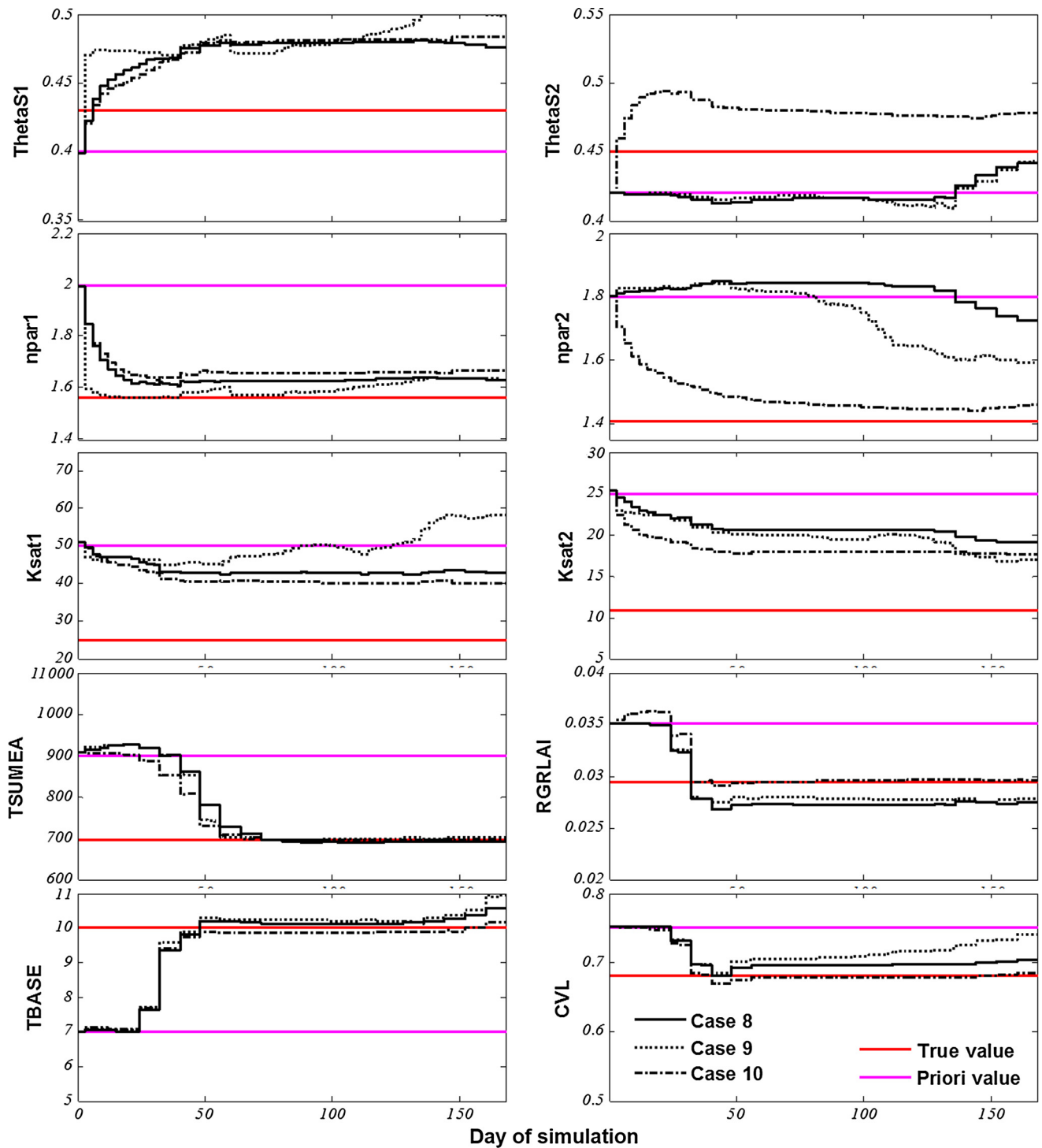


Fig. 11. The temporal evolution of estimated model parameters of Cases 8–10 over the simulation period.

whole simulation period (Fig. 11). For Case 8, the estimation of soil parameters in deep soil is bad. The stratified soil profile makes it very challenging to identify deep soil parameters due to the weak correlation between surface soil moisture and deep soil parameters. Similar findings have been reported by Lü et al. (2011) and Shi et al. (2015), which showed that surface soil moisture data might be capable of identifying soil parameters for the top layers, but exert little or no influence on the deeper soil. SSM observation is helpful in correcting the soil water content in the top layer, but bring very limited improvement to the deep soil moisture (Fig. 12).

The bad estimations of soil parameters and soil moisture of the second layer caused a few negative consequences. Firstly, the estimated water stress deviates significantly from the observation (Fig. 12). Secondly, comparing Case 8 with Case 5, the estimates of RGRLAI and CVL deteriorate in Case 8 (Fig. 10). Although the observation errors of LAI and SSM in Case 9 have been reduced, the improvements of soil and crop parameter, deep soil moisture profile and water stress estimation are not noticeable.

To improve the deep soil parameter estimation, deep soil observations such as deep soil moisture, soil water pressure head, and

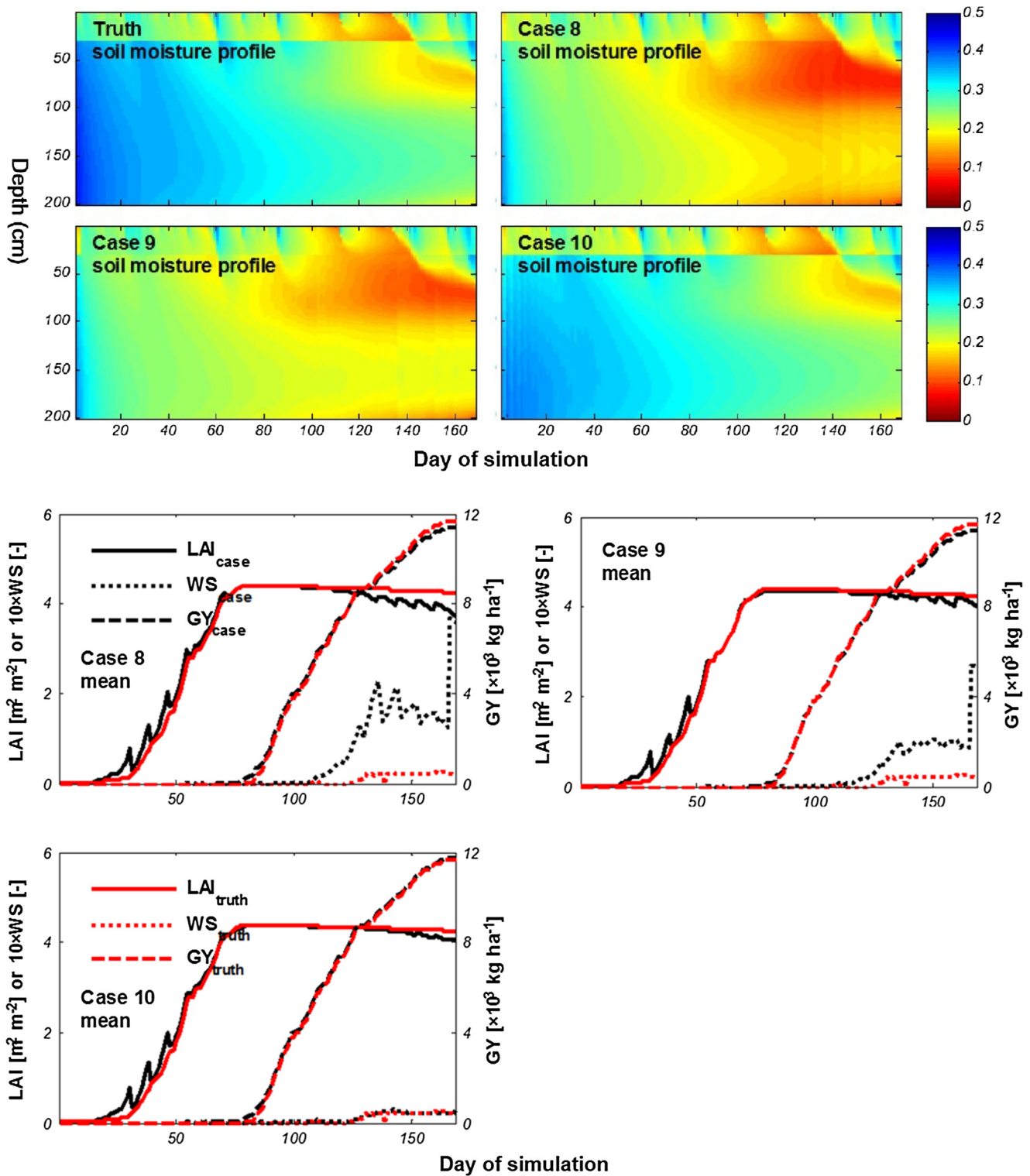


Fig. 12. The temporal evolution of soil moisture profile and crop outputs of cases 8–10.

groundwater level data will be necessary (Shi et al., 2015). Case 10 incorporates an additional point-scale soil moisture observation at a depth of 40 cm. An obvious improvement to npar2 estimation is achieved. At the same time, the estimation of crop parameters, soil water status, water stress, and crop growth and development is improved. We thus can conclude that, by only assimilating LAI and SSM observation, stratified soil profile not only increases the difficulty of accurately estimating soil parameters and the soil moisture profile but also challenges the estimation of crop param-

eters and status if crop needs to uptake deeper soil water. Reducing the observation errors of LAI and SSM is not beneficial, while extra deep soil moisture observation is helpful. It indicates that, during measurement design, since remote sensing measurements can only detect the spatial/temporal distribution of soil moisture in the top few centimeters of the soil surface, ground soil moisture sensor networks focusing on the deep information is required under soil stratification condition, especially when the crop growth depends on the deep soil water supply.

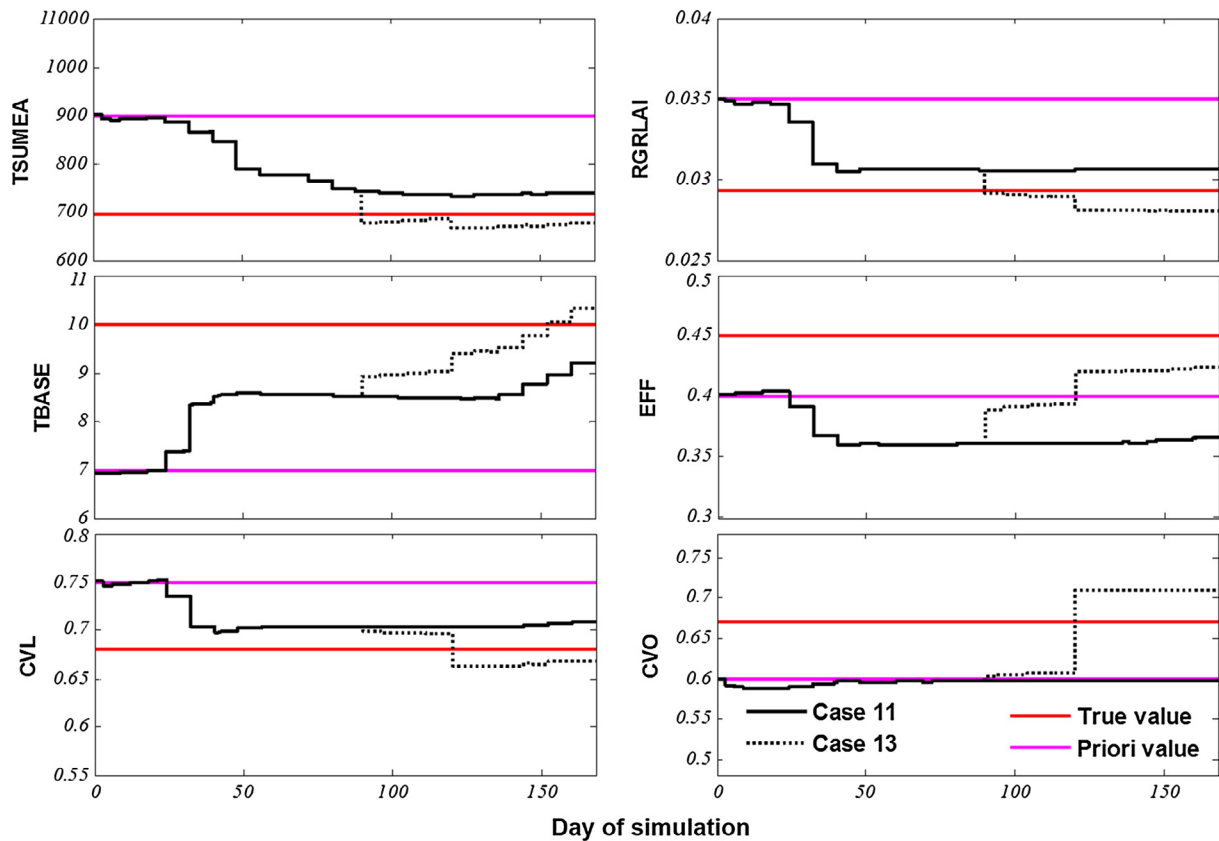


Fig. 13. The temporal evolution of estimated crop parameters of Cases 11 and 13 over the simulation period.

3.4. Handling an increasing number of unknown crop parameters

Based on Case 10, Cases 11–13 introduce two more crop parameters CVO and EFF (component P3 in Table 2). CVO is the efficiency of conversion into storage organs, which has a direct influence on

grain yield (GY) accumulation. EFF is the light-use efficiency for real leaf, which is an important parameter affecting the LAI development during the vegetative stage and the grain yield accumulation during the reproductive stage. There are totally 12 unknown parameters in these cases. Case 11 employs SSPE strategy,

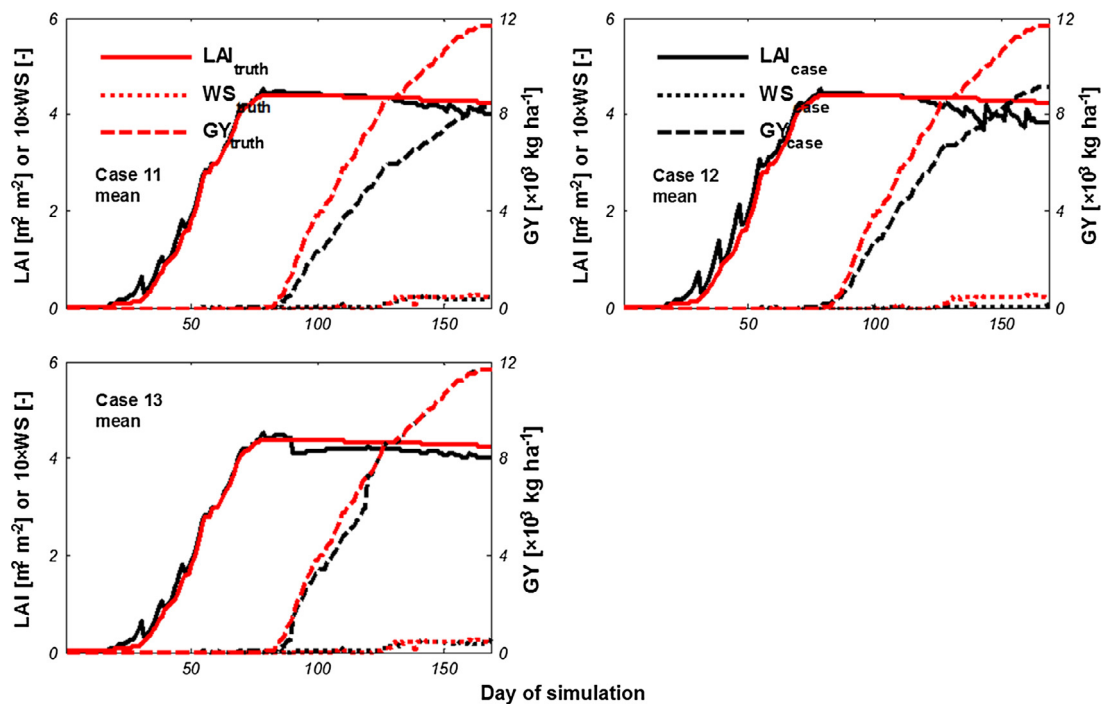


Fig. 14. The temporal evolution of crop growth outputs of Cases 11–13 over the simulation period.

comparing with Case 12 using USO strategy. The observations of LAI, SSM and SM40 are simultaneously assimilated in Case 11 and 12, while additional grain yield (GY) observation is also assimilated in Case 13. The grain yield is observed at $t = 120$ and 150 days with observation error variance of 100^2 kg ha^{-1} . Comparison between Case 13 and Case 11 is to examine the necessity of collecting more types of crop data if more crop parameters are poorly known. A summary of Cases 11–13 is presented in Table 3.

Only crop parameter estimates of Cases 11 and 13 are presented in Fig. 13, and the estimated LAI, grain yield, and water stress are shown in Fig. 14. Comparing the results of case 11 to case 10, it is found that the crop parameter estimation severely deteriorates. In Case 11, LAI development is influenced by more poorly known parameters (TSUMEA, RGRLAI, TBASE, CVL, and EFF) than in Case 10. It indicates that only the information content of LAI and soil moisture data is insufficient to simultaneously derive the correct estimates for the increasing number of poorly known crop parameters. Furthermore, because of the very low correlation between given observations (LAI, SSM, SM40) and parameter CVO (figures not presented), CVO does not show any updating in Case 11.

Although LAI has been reproduced very well in Fig. 14, the grain yield estimation in Case 11 has a significant difference with the reference value. Among these poorly known crop parameters, TBASE, EFF, and CVO are the three parameters affecting the grain yield accumulation, and the deviated grain yield estimate is attributed to the bad estimation of parameter TBASE, EFF, and CVO. The available observations cannot provide enough constraint for the estimation of these parameters. From Fig. 14, it is also interesting to notice that the obtained grain yield curve from USO strategy is closer to the reference than that from SSPE strategy. SSPE assimilation strategy is no longer advantageous in terms of grain yield prediction. The corrupted parameter estimates and deteriorated grain yield prediction in Case 11 highlight the non-necessity of running joint state-parameter data assimilation.

Fortunately, the estimation of TBASE, EFF, and CVO provides an obvious improvement if grain yield data is added to the observation set. The results indicate that in a system with many poorly known crop parameters, assimilating adequate and appropriate crop data will be necessary if simultaneous state-parameter estimation is employed. From the point of view of measurement design, far more data besides the most accessible LAI and SSM are required for a soil-water-atmosphere-plant system.

4. Conclusion

Through extensive OSS experiments, this study focused on two tasks: Task 1: evaluating how well the SWAP model can perform

with given measurements under some challenging conditions (e.g., different state vector construction, stratified soil profile, increasing number of poorly known crop parameters which are less sensitive to measurements); Task 2: if the model does not work well, providing the corresponding suggestions in measurement design to improve the estimation performance.

Our results showed that a proper selection of the state vector is critical to effective data assimilation. It demonstrated that, to accurately estimate soil water and crop growth, the state vector should include soil water content, the specific leaf area, the dry matter weight of leaves, the physiological age of leaves, class number of living leaves, and development stage. Especially, updating development stage was able to avoid the negative effect of “phenological shift”.

Our results showed that SSPE assimilation strategy, in some cases, can lead to accuracy improvement and uncertainty reduction, compared to USO assimilation strategy. An advantage of SSPE assimilation strategy was that it can alleviate the inconsistency between parameters and state variables. However, with an increasing number of uncertain parameters due to stratified soil profile and more poorly known crop parameters, the performance of SSPE assimilation strategy may deteriorate with corrupted parameter estimates. Besides the most easily available SSM and LAI measurements, deep soil moisture, grain yield or other auxiliary data were required to provide sufficient constraints on parameter estimation and to assure the data assimilation performance.

With the fast emergence of remote sensing technology, we anticipate the near real-time data acquisition with adequate temporal and spatial resolution and accuracy for agricultural applications. However, this study emphasizes that for real systems with soil stratification and a large number of unknown crop parameters, we need to pay further attention to acquiring and assimilating underground (for example, deep soil moisture) and aboveground (for example, crop yield) measurements.

Acknowledgements

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Appendix A

Table A1

Parameters and statistical setting used for sensitivity analysis (CV: coefficient of variance; [minimum, maximum]). The soil is loam. The statistics of soil parameters are defined as the mean and standard deviation (SD).

Parameter	Definition	Default parameter value	Perturbation (CV)	Carsel and Parrish (1988) (mean, SD)	Boons-Prins et al. (1993)	Ceglar et al. (2011)	Wang et al. (2013)
thetaR (θ_r)	Residual water content ($\text{cm}^3 \text{ cm}^{-3}$)	0.078	$\pm 10\%$	(0.078, 0.013)	–	–	–
thetaS (θ_s)	Saturated water content ($\text{cm}^3 \text{ cm}^{-3}$)	0.43	$\pm 10\%$	(0.43, 0.10)	–	–	–
Alfa (α)	Shape parameter alfa of main drying curve (cm^{-1})	0.036	$\pm 10\%$	(0.036, 0.021)	–	–	–
npar(n)	Shape parameter (–)	1.56	$\pm 10\%$	(1.56, 0.11)	–	–	–
Ksat(K_s)	Saturated vertical hydraulic conductivity (cm d^{-1})	24.96	$\pm 10\%$	(24.96, 43.58)	–	–	–
TSUMEA	Temperature sum from emergence to anthesis ($^{\circ}\text{C d}$)	695	$\pm 10\%$	–	[695,935]	–	–
TSUMAM	Temperature sum from anthesis to maturity ($^{\circ}\text{C d}$)	786	$\pm 10\%$	–	[800,920]	–	–

Table A1 (continued)

Parameter	Definition	Default parameter value	Perturbation (CV)	Carsel and Parrish (1988) (mean, SD)	Boons-Prins et al. (1993)	Ceglar et al. (2011)	Wang et al. (2013)
TDWI	Initial total crop dry weight (kg ha^{-1})	5	$\pm 10\%$	–	–	–	–
LAEM	Leaf area index at emergence ($\text{m}^{-2} \text{m}^{-2}$)	0.0074	$\pm 10\%$	–	–	[0.04,0.09]	[0.0435,0.0532]
RRLAI	Maximum relative increase in LAI ($\text{m}^{-2} \text{m}^{-2} \text{d}^{-1}$)	0.0294	$\pm 10\%$	–	–	[0.02,0.04]	[0.0265,0.0323]
SPAN	Life span of leaves growing at 35 °C (d)	31	$\pm 10\%$	–	–	[30,35]	[29.7,36.3]
TBASE	Lower threshold temperature for ageing of leaves (°C)	10	$\pm 10\%$	–	–	[8,10]	[9,11]
SLATB ₀	Specific leaf area (ha kg^{-1} , DVS = 0)	0.0035	$\pm 10\%$	–	–	[0.0022,0.0035]	[0.0023,0.0029]
SLATB _{0.78}	Specific leaf area (ha kg^{-1} , DVS = 0.78)	0.0016	$\pm 10\%$	–	–	[0.0010,0.0018]	[0.0011,0.0013]
SLATB _{2.0}	Specific leaf area (ha kg^{-1} , DVS = 2.0)	0.0016	$\pm 10\%$	–	–	[0.0010,0.0018]	[0.0011,0.0013]
EFF	Light-use efficiency for real leaf ($\text{kg CO}_2 \text{J}^{-1}$)	0.45	$\pm 10\%$	–	–	[0.45,0.55]	[0.405,0.495]
AMAXTB ₀	Max CO ₂ assimilation rate ($\text{kg ha}^{-1} \text{hr}^{-1}$, DVS = 0)	70	$\pm 10\%$	–	–	[65,72]	[63,77]
AMAXTB _{1.25}	Max CO ₂ assimilation rate ($\text{kg ha}^{-1} \text{hr}^{-1}$, DVS = 1.25)	70	$\pm 10\%$	–	–	[65,72]	–
AMAXTB _{1.5}	Max CO ₂ assimilation rate ($\text{kg ha}^{-1} \text{hr}^{-1}$, DVS = 1.5)	63	$\pm 10\%$	–	–	[55,65]	[56.7,69.3]
AMAXTB _{1.75}	Max CO ₂ assimilation rate ($\text{kg ha}^{-1} \text{hr}^{-1}$, DVS = 1.75)	49	$\pm 10\%$	–	–	[45,50]	[44.1,53.9]
AMAXTB _{2.0}	Max CO ₂ assimilation rate ($\text{kg ha}^{-1} \text{hr}^{-1}$, DVS = 2.0)	21	$\pm 10\%$	–	–	[15,25]	[18.9,23.1]
TMPFTB ₉	Reduction factor of AMAX (–, AveT = 9 °C)	0.05	$\pm 10\%$	–	–	[0.05,0.225]	[0.045,0.055]
TMPFTB ₁₆	Reduction factor of AMAX (–, AveT = 16 °C)	0.8	$\pm 10\%$	–	–	[0.48,0.80]	[0.72,0.88]
TMPFTB ₁₈	Reduction factor of AMAX (–, AveT = 18 °C)	0.94	$\pm 10\%$	–	–	[0.55,0.94]	[0.85,1.00]
TMPFTB ₂₀	Reduction factor of AMAX (–, AveT = 20 °C)	1	$\pm 10\%$	–	–	[0.63,1.00]	[0.90,1.00]
CVL	Efficiency of conversion into leaves (kg kg^{-1})	0.68	$\pm 10\%$	–	–	[0.68,0.72]	[0.61,0.75]
CVO	Efficiency of conversion into storage organs (kg kg^{-1})	0.671	$\pm 10\%$	–	–	[0.73,0.76]	[0.60,0.74]
CVR	Efficiency of conversion into roots (kg kg^{-1})	0.69	$\pm 10\%$	–	–	[0.65,0.79]	[0.62,0.76]
CVS	Efficiency of conversion into stems (kg kg^{-1})	0.658	$\pm 10\%$	–	–	[0.65,0.72]	[0.59,0.72]
Q ₁₀	Relative increase in respiration rate per 10 °C temperature increase (–)	2	$\pm 10\%$	–	–	[1.5,2.0]	[1.8,2.2]
RML	Relative maintenance respiration rate of leaves ($\text{kgCH}_2\text{O kg}^{-1} \text{d}^{-1}$)	0.03	$\pm 10\%$	–	–	[0.003,0.011]	[0.027,0.033]
RMO	Relative maintenance respiration rate of storage organs ($\text{kgCH}_2\text{O kg}^{-1} \text{d}^{-1}$)	0.01	$\pm 10\%$	–	–	[0.005,0.010]	[0.009,0.011]
RMR	Relative maintenance respiration rate of roots ($\text{kgCH}_2\text{O kg}^{-1} \text{d}^{-1}$)	0.015	$\pm 10\%$	–	–	[0.006,0.010]	[0.014,0.017]
RMS	Relative maintenance respiration rate of stems ($\text{kgCH}_2\text{O kg}^{-1} \text{d}^{-1}$)	0.015	$\pm 10\%$	–	–	[0.006,0.010]	[0.014,0.017]
RFSETB _{1.75}	Reduction factor of senescence (–, DVS = 1.75)	0.75	$\pm 10\%$	–	–	[0.70,0.80]	[0.23,0.68]
RFSETB _{2.0}	Reduction factor of senescence (–, DVS = 2.0)	0.25	$\pm 10\%$	–	–	[0.20,0.30]	[0.28,0.63]
FRTB ₀	Fraction of total dry matter to the roots (–, DVS = 0)	0.4	$\pm 10\%$	–	–	[0.35,0.40]	[0.36,0.44]
FRTB _{0.4}	Fraction of total dry matter to the roots (–, DVS = 0.4)	0.27	$\pm 10\%$	–	–	[0.25,0.30]	[0.34,0.30]
FRTB _{0.6}	Fraction of total dry matter to the roots (–, DVS = 0.6)	0.19	$\pm 10\%$	–	–	[0.19,0.23]	[0.17,0.21]
FRTB _{0.9}	Fraction of total dry matter to the roots (–, DVS = 0.9)	0.06	$\pm 10\%$	–	–	[0.06,0.10]	[0.05,0.07]
FLTB ₀	Fraction of aboveground dry matter to the leaves (–, DVS = 0)	0.7	$\pm 10\%$	–	–	[0.55,0.65]	[0.56,0.68]
FLTB _{0.33}	Fraction of aboveground dry matter to the leaves (–, DVS = 0.33)	0.7	$\pm 10\%$	–	–	[0.55,0.63]	[0.56,0.68]
FLTB _{0.88}	Fraction of aboveground dry matter to the leaves (–, DVS = 0.88)	0.15	$\pm 10\%$	–	–	[0.10,0.20]	[0.14,0.17]
FSTB ₀	Fraction of aboveground dry matter to the stems (–, DVS = 0)	0.3	$\pm 10\%$	–	–	[0.35,0.45]	[0.32,0.44]
FSTB _{0.33}	Fraction of aboveground dry matter to the stems (–, DVS = 0.33)	0.3	$\pm 10\%$	–	–	[0.37,0.45]	[0.32,0.44]
FSTB _{0.88}	Fraction of aboveground dry matter to the stems (–, DVS = 0.88)	0.85	$\pm 10\%$	–	–	[0.80,0.90]	[0.83,0.86]
FOTB _{1.05}	Fraction of aboveground dry matter to the storage organs (–, DVS = 1.05)	1	$\pm 10\%$	–	–	[0.90,1.0]	[0.9,1.0]
FOTB _{2.0}	Fraction of aboveground dry matter to the storage organs (–, DVS = 2.0)	1	$\pm 10\%$	–	–	[0.90,1.0]	[0.9,1.0]
RDRRTB _{1.5001}	Relative death rates of roots (–, DVS = 1.5001)	0.02	$\pm 10\%$	–	–	[0.005,0.020]	[0.018,0.022]
RDRRTB _{2.0}	Relative death rates of roots (–, DVS = 2.0)	0.02	$\pm 10\%$	–	–	[0.005,0.020]	[0.018,0.022]
RDRSTB _{1.5001}	Relative death rates of stems (–, DVS = 1.5001)	0.02	$\pm 10\%$	–	–	–	–
RDRSTB _{2.0}	Relative death rates of stems (–, DVS = 2.0)	0.02	$\pm 10\%$	–	–	–	–
COFAB	Interception coefficient (cm)	0.25	$\pm 10\%$	–	–	–	–
RDI	Initial rooting depth (cm)	10	$\pm 10\%$	–	–	[7,10]	[9,11]
RRI	Maximum daily increase in rooting depth (cm d^{-1})	1.5	$\pm 10\%$	–	–	[1.5,3.0]	[1.98,2.42]
RDC	Maximum rooting depth crop/cultivar (cm)	100	$\pm 10\%$	–	–	[80,130]	[90,110]

References

- Andrea, S.K., Yohannes, T.Y., Vristine, L.S.M., Bright, C.D., 2017. Evaluation and improvement of the default soil hydraulic parameters for the Noah Land Surface Model. *Geoderma* 285, 247–259.
- Barthélémy, S., Ricci, S., Rochoux, M.C., Le Pape, E., Thual, O., 2017. Ensemble-based data assimilation for operational flood forecasting – On the merits of state estimation for 1 D hydrodynamic forecasting through the example of the “Adour Maritime” river. *J. Hydrol.* 552, 210–224.
- Bloom, A.A., Williams, M., 2015. Constraining ecosystem carbon dynamics in a data-limited world: integrating ecological “common sense” in a model-data-fusion framework. *Biogeosciences* 12, 1299–1315.
- Boons-Prins, E.R., de Koning, G.H.J., van Diepen, C.A., Penning De Vries, F.W.T., 1993. Crop-specific simulation parameters for yield forecasting across the European Community. CABO-DLO, 6700 AA Wageningen, The Netherlands.

- Bouten, W., 1992. Monitoring and modelling forest hydrological processes in support of acidification research. Univ., A'dam, Diss.
- Brisson, N., Gary, C., Justes, E., Roche, R., Mary, B., Ripoche, D., Zimmer, D., Sierra, J., Bertuzzi, P., Bussi re, F., Cabidoche, Y.M., Cellier, P., Debaeke, P., Gaudill re, J.P., Henault, C., Maraux, F., Seguin, B., Sinoquet, H., 2003. An overview of the crop model STICS. *Europ. J. Agronomy* 18, 309–332.
- Campolongo, F., Carboni, B., Saltelli, A., Schoutens, W., 2004. Enhancing the Morris method. In: Kenneth M. Hanson and Fran ois M. Hemez, eds. *Proceedings of the 4th International Conference on Sensitivity Analysis of Model Output*. Santa Fe, New Mexico, 369–379.
- Carsel, R.F., Parrish, R.S., 1988. Developing joint probability distribution of soil water retention characteristics. *Water Resour. Res.* 24, 755–769.
- Ceglar, A.,  repi sek, Z., Kajfe z-Bogataj, L., Poga ar, T., 2011. The simulation of phenological development in dynamic crop model: The Bayesian comparison of different methods. *Agric. For. Meteorol.* 151, 101–115.
- Chen, W., Huang, C., Shen, H., Li, X., 2015. Comparison of ensemble-based state and parameter estimation methods for soil moisture data assimilation. *Adv. Water Resour.* 86, 425–438.
- Claverie, M., Demarez, V., Duchemin, B., Hagolle, O., Ducrot, D., Marais-Sicre, C., Dejoux, J., Huc, M., Keravec, P., Beziat, P., Fieuzal, R., Ceschia, E., Dedieu, G., 2012. Maize and sunflower biomass estimation in southwest France using high spatial and temporal resolution remote sensing data. *Remote Sens. Environ.* 124, 844–857.
- Crow, W., Kustas, W., Prueger, J., 2008. Monitoring root-zone soil moisture through the assimilation of a thermal remote sensing-based soil moisture proxy into a water balance model. *Remote Sens. Environ.* 112 (4), 1268–1281.
- Curnel, Y., de Wit, A.J.W., Duveiller, G., Defourny, P., 2011. Potential performances of remotely sensed LAI assimilation in WOFOST model based on an OSS Experiment. *Agric. For. Meteorol.* 151 (12), 1843–1855.
- Das, N.N., Mohanty, B.P., Cosh, M.H., Jackson, T.J., 2008. Modeling and assimilation of root zone soil moisture using remote sensing observations in Walnut Gulch Watershed during SMEX04. *Remote Sens. Environ.* 112 (2), 415–429.
- de Wit, A., Duveiller, G., Defourny, P., 2012. Estimating regional winter wheat yield with WOFOST through the assimilation of green area index retrieved from MODIS observations. *Agric. For. Meteorol.* 164, 39–52.
- de Wit, A., van Diepen, C.A., 2007. Crop model data assimilation with the Ensemble Kalman filter for improving regional crop yield forecasts. *Agric. For. Meteorol.* 146 (1–2), 38–56.
- Dente, L., Satalino, G., Mattia, F., Rinaldi, M., 2008. Assimilation of leaf area index derived from ASAR and MERIS data into CERES-Wheat model to map wheat yield. *Remote Sens. Environ.* 112 (4), 1395–1407.
- Dorigo, W.A., Zurita-Milla, R., de Wit, A., Brazile, J., Singh, R., Schaepman, M.E., 2007. A review on reflective remote sensing and data assimilation techniques for enhanced agroecosystem modeling. *Int. J. Appl. Earth Obs. Geoinf.* 9 (2), 165–193.
- Duan, D., Gupta, V.K., Sorooshian, S., 1993. A shuffled complex evolution approach for effective and efficient global minimization. *J. Optim. Theory Appl.* 76 (3), 501–521.
- Entekhabi, D., Nakamura, H., Njoku, E.G., 1994. Solving the inverse problem for soil moisture and temperature profiles by sequential assimilation of multifrequency remotely sensed observations. *IEEE Trans. Geosci. Remote Sens.* 32 (2), 438–448.
- Evensen, G., 1994. Sequential data assimilation with a nonlinear quasi-geostrophic model using Monte Carlo methods to forecast error statistics. *J. Geophys. Res.* 99, 10143–10162.
- Feddes, R.A., Kowalik, P.J., Zaradny, H., 1978. *Simulation of field water use and crop yield*. Centre for Agricultural Publishing and Documentation, Wageningen, Netherlands.
- Freer, J., Beven, K.J., Ambrose, B., 1996. Bayesian estimation of uncertainty in runoff prediction and the value of data: An application of the GLUE approach. *Water Resour. Res.* 32, 2161–2173.
- Gan, Y.J., Duan, Q.Y., Gong, W., Tong, C., Sun, Y.W., Chu, W., Ye, A.Z., Miao, C.Y., Di, Z. H., 2014. A comprehensive evaluation of various sensitivity analysis methods: a case study with a hydrological model. *Environmental Modelling & Software*, 51, 269–285.
- Ge, Q., Menendez, M., 2017. Extending Morris method for qualitative global sensitivity analysis of models with dependent inputs. *Reliability Engineering and System Safety*, 162, 28–39.
- Haboudane, D., 2004. Hyperspectral vegetation indices and novel algorithms for predicting green LAI of crop canopies: Modeling and validation in the context of precision agriculture. *Remote Sens. Environ.* 90 (3), 337–352.
- Hansen, J.W., Jones, J.W., 2000. Scaling-up crop models for climate variability applications. *Agric. Syst.* 65 (1), 43–72.
- He, L., Song, X., Feng, W., Guo, B., Zhang, Y., Wang, Y., Wang, C., Guo, T., 2016. Improved remote sensing of leaf nitrogen concentration in winter wheat using multi-angular hyperspectral data. *Remote Sens. Environ.* 174, 122–133.
- Hoogenboom, G., 2004. *Decision Support System for Agrotechnology Transfer, Version 4.0*, Technical Report. Univ. of Hawaii, Honolulu.
- Huang, C., Chen, W., Li, Y., Shen, H., Li, X., 2016. Assimilating multi-source data into land surface model to simultaneously improve estimations of soil moisture, soil temperature, and surface turbulent fluxes in irrigated fields. *Agric. For. Meteorol.* 230, 142–156.
- Huang, J., Tian, L., Liang, S., Ma, H., Becker-Reshef, I., Huang, Y., Su, W., Zhang, X., Zhu, D., Wu, W., 2015. Improving winter wheat yield estimation by assimilation of the leaf area index from Landsat TM and MODIS data into the WOFOST model. *Agric. For. Meteorol.* 204, 106–121.
- Huang, Y., Zhu, Y., Li, W.L., Cao, W.X., Tian, Y.C., 2013. Assimilating Remotely Sensed Information with the WheatGrow Model Based on the Ensemble Square Root Filter for Improving Regional Wheat Yield Forecasts. *Plant Prod Sci* 16 (4), 352–364.
- Ines, A.V.M., Das, N.N., Hansen, J.W., Njoku, E.G., 2013. Assimilation of remotely sensed soil moisture and vegetation with a crop simulation model for maize yield prediction. *Remote Sens. Environ.* 138, 149–164.
- Keating, E.H., Doherty, J., Vrugt, J.A., Kang, Q., 2010. Optimization and uncertainty assessment of strongly nonlinear groundwater models with high parameter dimensionality. *Water Resour. Res.* 46 (10).
- King, D.M., Perera, B.J.C., 2013. Morris method of sensitivity analysis applied to assess the importance of input variables on urban water supply yield-A case study. *J. Hydrol.* 477, 17–32.
- Lakhankar, T., Krakauer, N., Khanbilvardi, R., 2009. Applications of microwave remote sensing of soil moisture for agricultural applications. *Int. J. Terraspace Sci. Eng.* 2 (1), 81–91.
- L , H., Yu, Z., Zhu, Y., Drake, S., Hao, Z., Sudicky, E., 2011. Dual state-parameter estimation of root zone soil moisture by optimal parameter estimation and extended Kalman filter data assimilation. *Adv. Water Resour.* 34 (3), 395–406.
- Ma, H., Huang, J., Zhu, D., Liu, J., Su, W., Zhang, C., Fan, J., 2013b. Estimating regional winter wheat yield by assimilation of time series of HJ-1 CCD NDMI into WOFOST-ACRM model with Ensemble Kalman Filter. *Math. Comput. Model.* 58 (3–4), 753–764.
- Ma, G., Huang, J., Wu, W., Fan, J., Zou, J., Wu, S., 2013. Assimilation of MODIS-LAI into the WOFOST model for forecasting regional winter wheat yield. *Math. Comput. Modell.*, 58(3–4), 634–643.
- MacDonald, R.B., Hall, F.G., 1980. *Global Crop Forecasting*. Science 208 (4445), 670–679.
- Meng, S.S., Xie, X.H., Liang, S.L., 2017. Assimilation of soil moisture and streamflow observations to improve flood forecasting with considering runoff routing lags. *J. Hydrol.* 550, 568–579.
- Monsivais-Huertero, A., Graham, W.D., Judge, J., Agrwal, D., 2010. Effect of simultaneous state-parameter estimation and forcing uncertainties on root-zone soil moisture for dynamic vegetation using EnKF. *Adv. Water Resour.* 33 (4), 468–484.
- Monteith, J.L., 1981. Evaporation and surface temperature. *Quarterly J. Royal Soc.* 107, 1–27.
- Montzka, C., Moradkhani, H., Weihermuller, L., Franssen, H., Canty, M., Vereecken, H., 2011. Hydraulic parameter estimation by remotely-sensed top soil moisture observations with the particle filter. *J. Hydrol.* 399 (3–4), 410–421.
- Moradkhani, H., Sorooshian, S., Gupta, H.V., Houser, P.R., 2005. Dual state-parameter estimation of hydrological models using ensemble Kalman filter. *Adv. Water Resour.* 28 (2), 135–147.
- Morris, M.D., 1991. Factorial sampling plans for preliminary computational experiments. *33(2)*, 161–174.
- Mualem, Y., 1976. A new model for predicting the hydraulic conductivity. *Water Resour. Res.* 12 (3), 513–522.
- Mutiga, J.K., Su, Z., Woldai, T., 2010. Using satellite remote sensing to assess evapotranspiration: Case study of the upper Ewaso Ng'iro North Basin, Kenya. *Int. J. Appl. Earth Obs. Geoinf.* 12, S100–S108.
- Nearing, G.S., Crow, W., Thorp, K., Moran, M., Reichle, R., Gupta, H., 2012. Assimilating remote sensing observations of leaf area index and soil moisture for wheat yield estimates: An observing system simulation experiment. *Water Resour. Res.* 48 (5), 213–223.
- Pathiraja, S., Marshall, L., Sharma, A., Morakhani, H., 2016. Detecting non-stationary hydrologic model parameters in a paired catchment system using data assimilation. *Adv. Water Resour.* 94, 103–119.
- Schoups, G., Vrugt, J.A., 2010. A formal likelihood function for parameter and predictive inference of hydrologic models with correlated, heteroscedastic, and non-Gaussian errors. *Water Resour. Res.* 46, 1–17.
- Sharpley, A.N., Williams, J.R., 1990. *EPIC Erosion/Productivity Impact Calculator*. US Dept. of Agriculture, Agricultural Research Service.
- Shi, L., Song, X., Tong, J., Zhu, Y., Zhang, Q., 2015. Impacts of different types of measurements on estimating unsaturated flow parameters. *J. Hydrol.* 524, 549–561.
- Skaggs, T.H., van Genuchten, M.T., Shouse, P.J., Poss, J.A., 2006. Macroscopic approaches to root water uptake as a function of water and salinity stress. *Agric. Water Manag.* 86 (1–2), 140–149.
- Song, X., Shi, L., Ye, M., Yang, J., Navon, I.M., 2014. Numerical comparison of iterative ensemble Kalman filters for unsaturated flow inverse modeling. *Vadose Zone J.* 13 (2), 1–12.
- Supit, I., Hooijer, A.A., Diepen, C.A., 1994. *System description of The WOFOST 6.0 Crop Simulation Model Implemented In CGMS, Volume 1: Theory and Algorithms*.
- Sus, O., Williams, M., Bernhofer, C., Beziat, P., Buchmann, N., Ceschia, E., Doherty, R., Eugster, W., Gruenwald, T., Kutsch, W., Smith, P., Wattenbach, M., 2010. A linked carbon cycle and crop developmental model: description and evaluation against measurements of carbon fluxes and carbon stocks at several European agricultural sites. *Agric. Ecosyst. Environ.* 139, 402–418.
- Sus, O., Poyatos, R., Barba, J., Carvalhais, N., Llorens, P., Williams, M., Vilalta, J., 2014. Time variable hydraulic parameters improve the performance of a mechanistic stand transpiration model. A case study of Mediterranean Scots pine sap flow data assimilation. *Agric. For. Meteorol.* 198–199, 168–180.
- van Dam, J.C. et al., 1997. *Theory of SWAP version 2.0*. Dep. Water Resour., Wageningen Agric. Univ., Wageningen, the Netherlands.

- van Genuchten, M.T., 1980. A closed-form equation for predicting the hydraulic conductivity of saturated soils. *SOIL SCI SOC AM J* 44 (5), 89–898.
- Vazifedoust, M., Van Dam, J.C., Bastiaanssen, W.G.M., Feddes, R.A., 2009. Assimilation of satellite data into agrohydrological models to improve crop yield forecasts. *Int. J. Remote Sens.* 30 (10), 2523–2545.
- Vrugt, J.A., ter Braak, C.J.F., Diks, C.G.H., Higdon, D., Robinson, B.A., Hyman, J.M., 2009. Accelerating Markov chain Monte Carlo simulation by differential evolution with self-adaptive randomized subspace sampling. *Int. J. Nonlinear Sci. Numer. Simulation.* 10 (3), 273–290.
- Vrugt, J.A., Diks, C.G.H., Gupta, H.V., Bouten, W., Verstraten, J.M., 2005. Improved treatment of uncertainty in hydrologic modeling: Combining the strengths of global optimization and data assimilation. *Water Resour. Res.* 41, W01017. <https://doi.org/10.1029/2004WR003059>.
- Vrugt, J.A., ter Braak, C.J.F., Clark, M.P., Hyman, J.M., Robinson, B.A., 2008. Treatment of input uncertainty in hydrologic modeling: Doing hydrology backward with Markov chain Monte Carlo simulation. *Water Resour. Res.* 44 (12), 5121–5127.
- Wang, J., Li, X., Lu, L., Fang, F., 2013. Parameter sensitivity analysis of crop growth models based on the extended Fourier Amplitude Sensitivity Test method. *Environ. Modell. Soft.* 48, 171–182.
- Yu, Z.B., Liu, D., Lü, H.S., Fu, X.L., Xiang, L., Zhu, Y.H., 2012. A multi-layer soil moisture data assimilation using support vector machines and ensemble particle filter. *J. Hydrol.* 475, 53–64.
- Zhang, Y., Chen, J.M., Miller, J.R., Noland, T.L., 2008. Leaf chlorophyll content retrieval from airborne hyperspectral remote sensing imagery. *Remote Sens. Environ.* 112 (7), 3234–3247.
- Zhang, H.J., Hendricks-Franssen, H.J., Han, X.J., Vrugt, J., Vereecken, H., 2016. Joint state and parameter estimation of two land surface models using the ensemble Kalman filter and particle filter. *Hydrol. Earth Syst. Sci. Discuss.* 42, 1–39.
- Zou, L., Zhan, C.S., Xia, J., Wang, T.J., Gippel, C.J., 2017. Implementation of evapotranspiration data assimilation with catchment scale distributed hydrological model via an ensemble Kalman filter. *J. Hydrol.* 549, 685–702.